

PANDA serial PG2

User Manual



Version 3.0

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1. Device User Guide for Windows System

1.1. Device Connection and Startup

1.1.1 Install Software Tool

Download Link: <https://github.com/SENSING-Technology/SensingCaptureV3>

Copy the SensingCaptureV3 installation package to the PC, unzip it, locate SensingCaptureV3.exe, and double-click it to open. Once opened, the interface will appear as shown in Figure 1-1. The control panel is at the top and left side, while the display area is in the center.

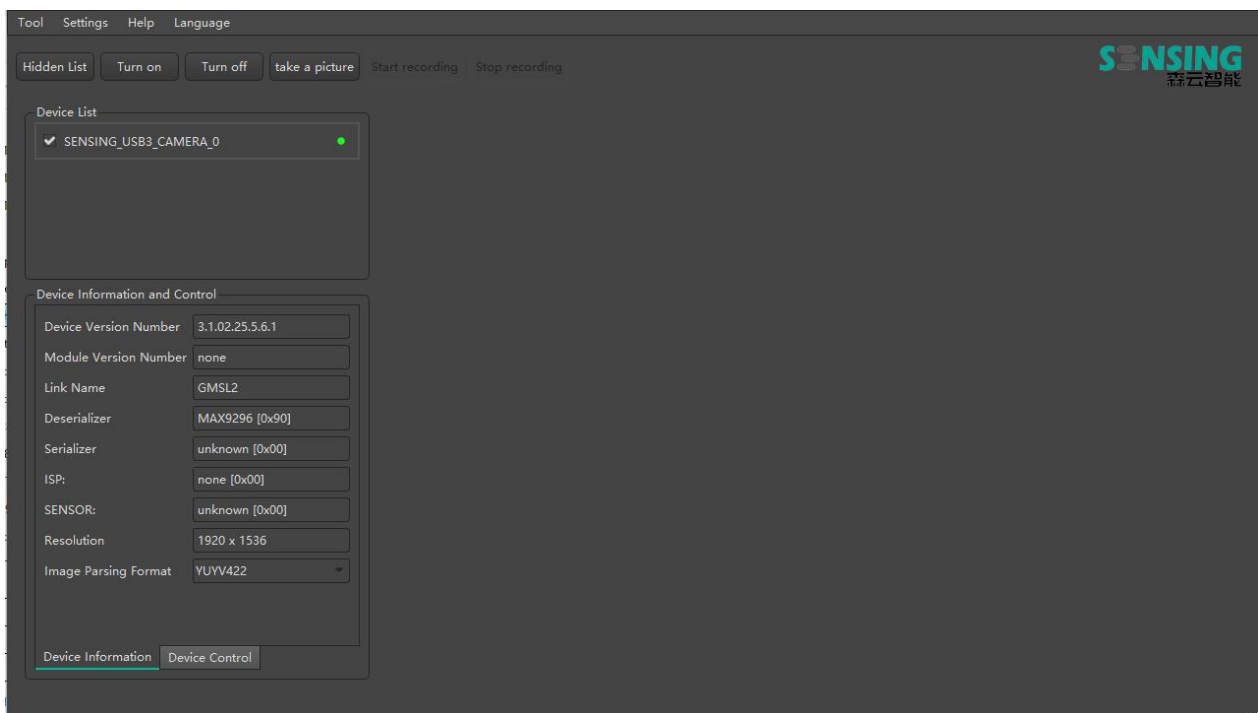
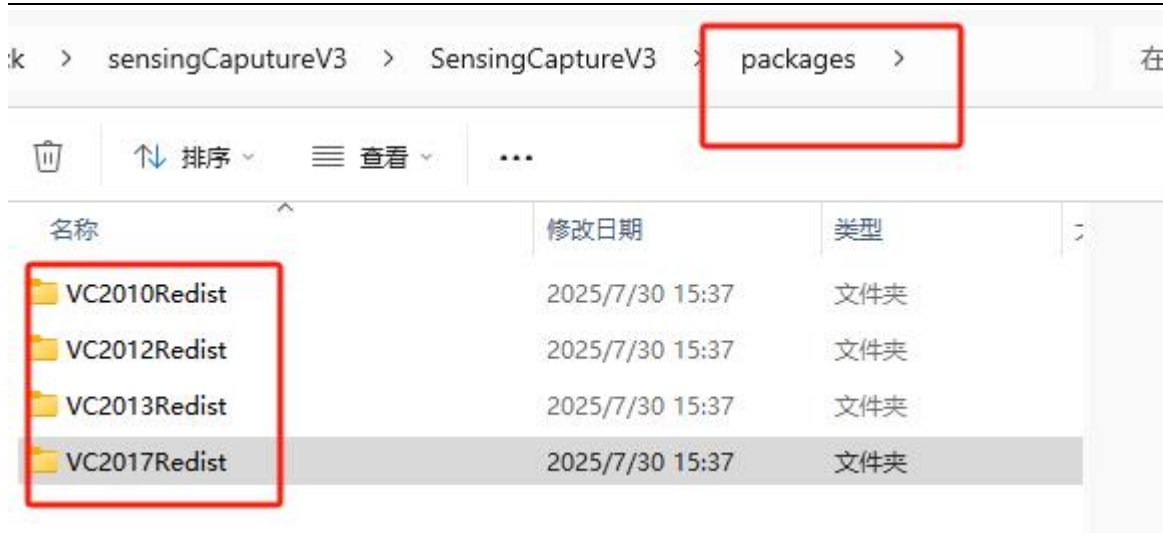


Figure 1-1

If this is the first time using the software and a prompt appears indicating that some library files are missing, you can try installing the VC201XRedist program found in the "packages" folder within the software directory.



1.1.2 Device Connection

Connect the device wiring according to Figure 1-2. The coaxial cable should be connected to the camera. When connecting one camera, the cable should be plugged into the CAM port.

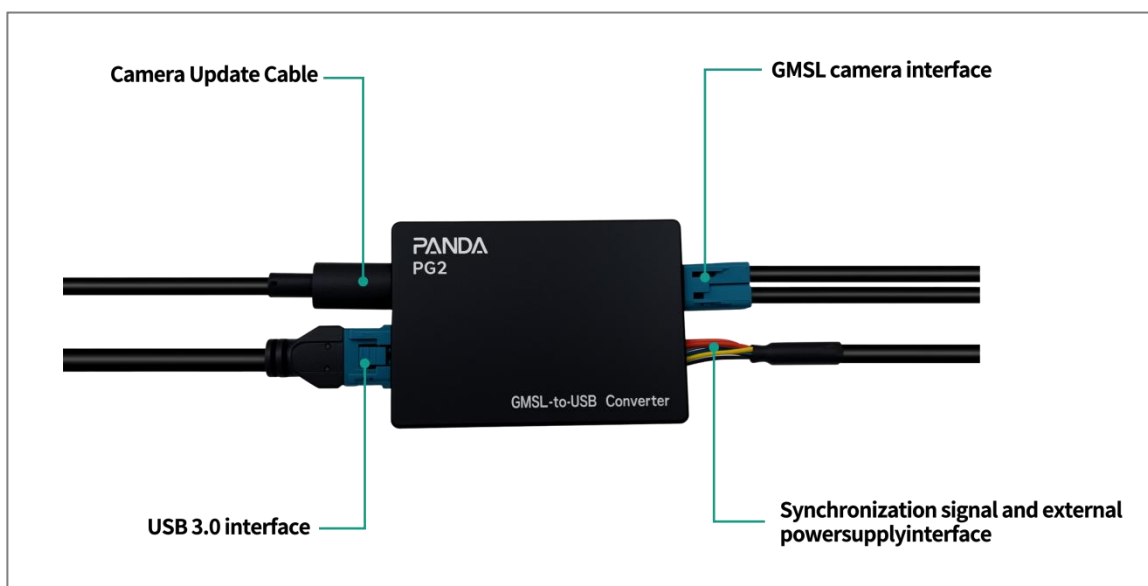


Figure 1-2

1.1.3 Enable the camera to capture images.

a. Open SensingCaptureV3

In the device list, you can see the camera device. At the same time, the current camera information can be viewed in the device signal and control panel.

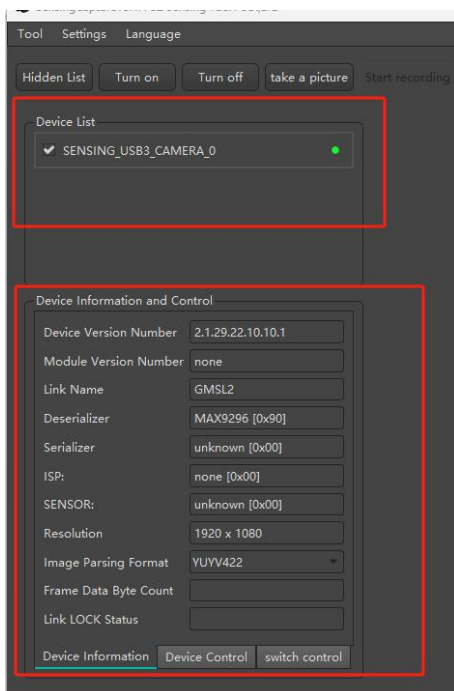


Figure 1-3

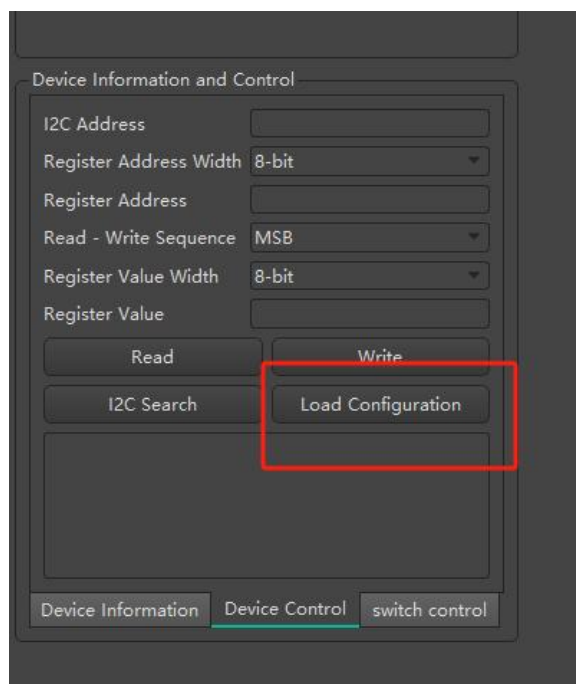


Figure 1-4

b. Load configuration

In the device information and control interface, click 'Device Control' and select 'Load Configuration.' A prompt box will appear for selecting the configuration file, as shown in Figure 1-4. There are two ways to load a configuration file:

Method 1:

In the configuration file field, click the small triangle and select the configuration file corresponding to the camera model from the dropdown menu, as shown in Figure 1-5

Method 2:

Use the fuzzy search feature to input part of the configuration file name. Based on the naming rule for configuration files, which follows the format: ProductModel_Resolution@FrameRate_ImageFormat, input partial information, such as ISX031, as shown in Figure 1-6.

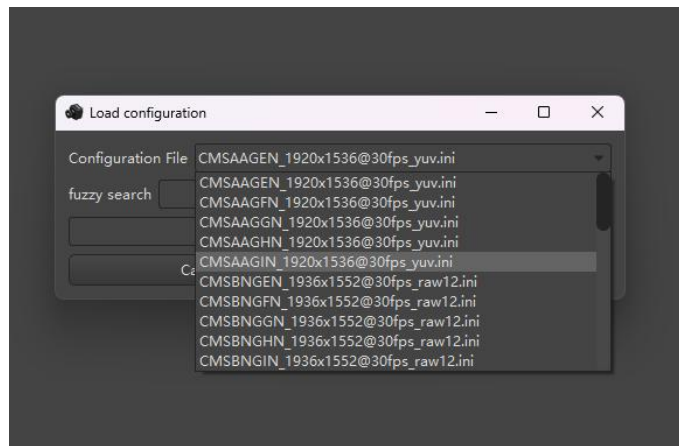


Figure 1-5

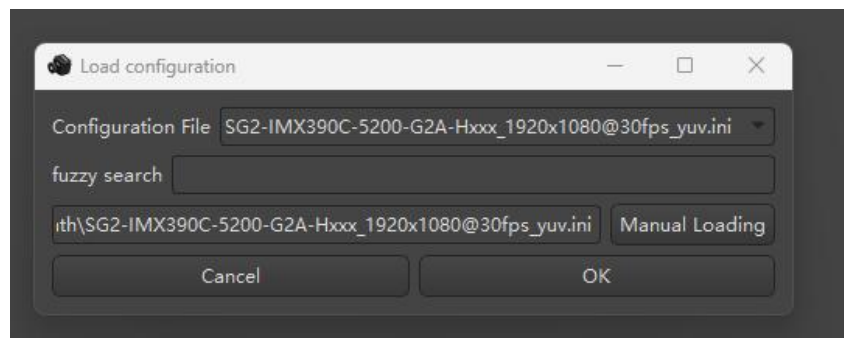


Figure 1-6

After selecting the configuration file, click the 'OK' button. The system will start the loading process based on the detailed information of the selected configuration file. Once the configuration file is successfully loaded, the system will display the information as shown in Figure 1-7.

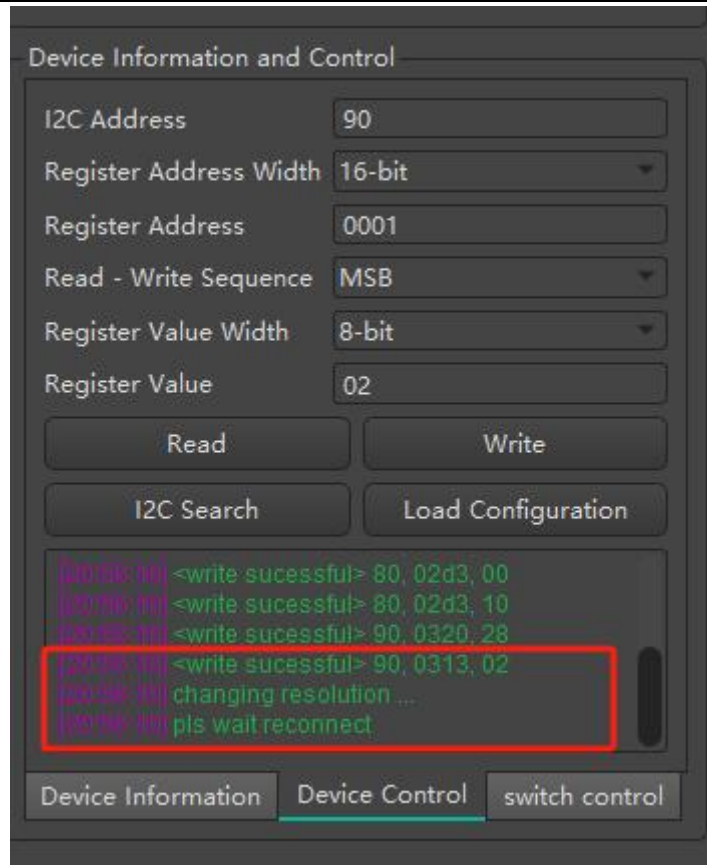


Figure 1-7

c. Open the camera.

Click on the 'Turn on' button in the top-left corner of the interface. The software interface will display the image captured by the camera in the middle of the screen, as shown in Figure 1-8.

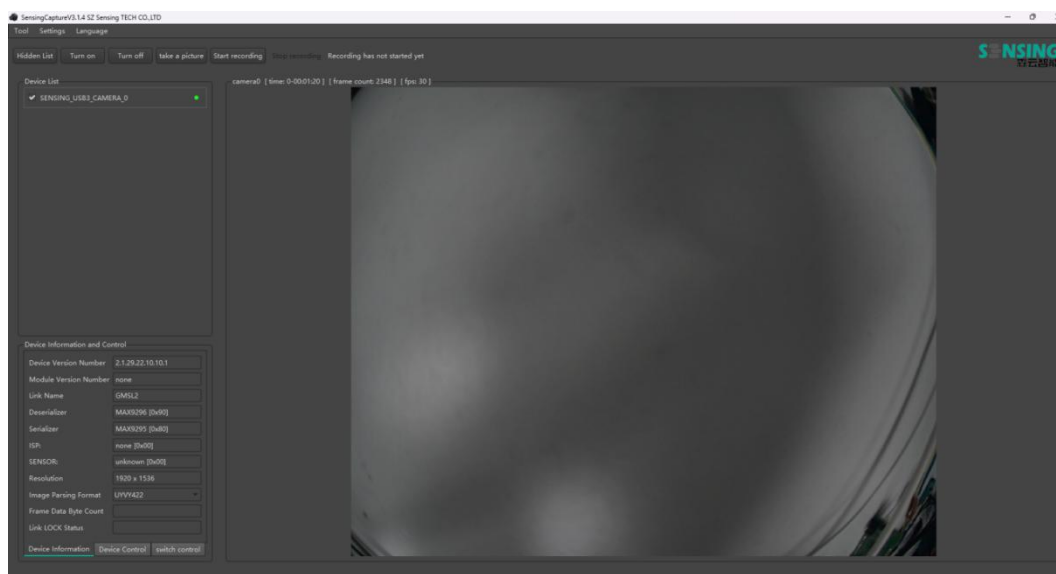


Figure 1-8

1.2 Function Description

1.2.1 Recording Function

A. Click the "Start Recording" button

Open the camera, and the interface will display as shown in Figure 1-9. Click the "Start Recording" button, and the system will pop up a window to choose the video save path. Based on personal needs, create an appropriate save path. During this process, a "Recording" prompt will appear, indicating that the recording is in progress. The recorded video will have a filename based on the time, with the file type being AVI (Audio Video Interleave) and the extension .avi. The video format will be Motion-JPEG (MJPEG).

B. Stop Recording

Click the "Stop Recording" button, and the interface will display as shown in Figure 1-10. This will stop the recording. The recorded video can then be found in the saved path you set earlier.

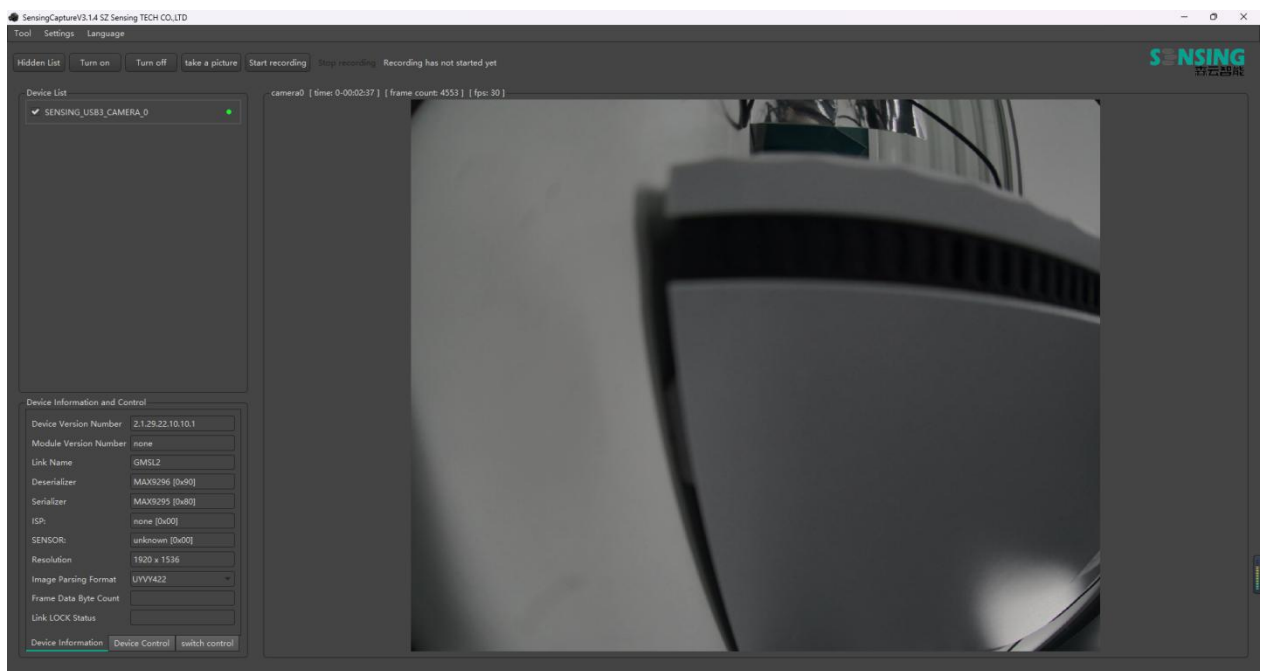


Figure 1-9

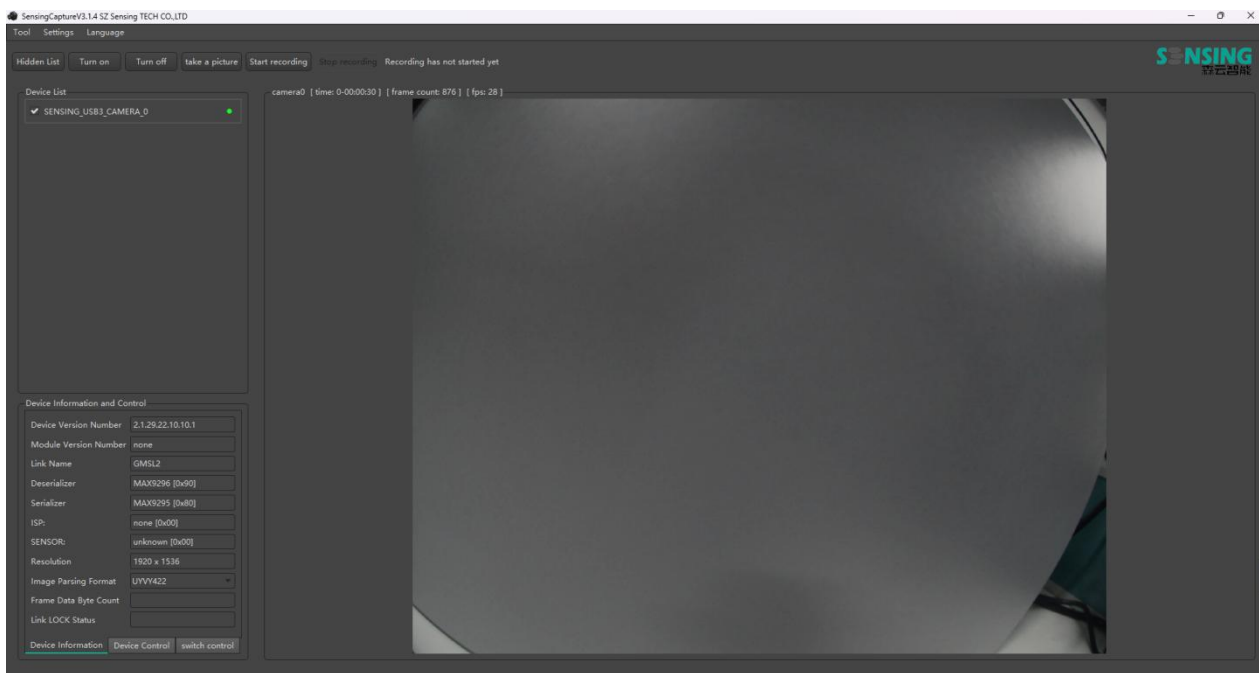


Figure 1-10

1.2.2 Photo function

Open the camera, click "Take a Picture," and the interface will display as shown in Figure 1-11. You can then take the picture. The file will be saved in the "pictures" directory under the software installation directory, as shown in Figure 1-12.

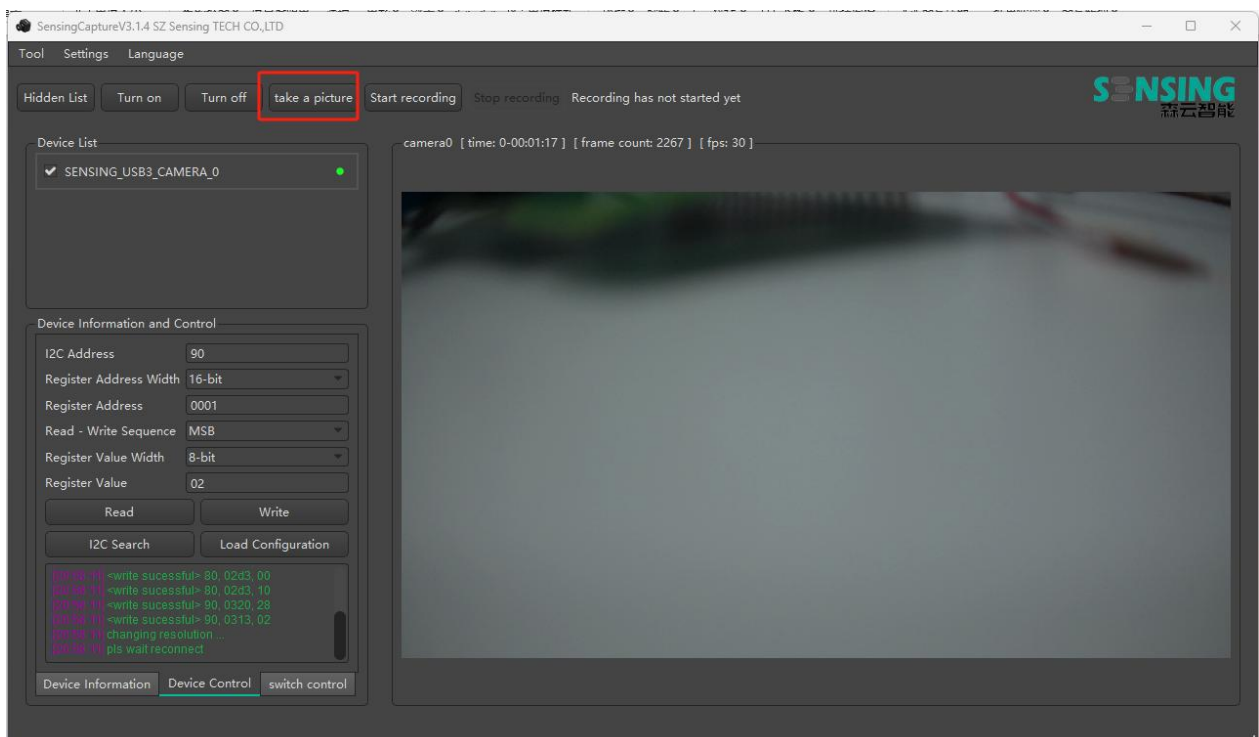


Figure 1-11

名称	修改日期	类型
bearer	2025/3/14 15:50	文件夹
configs	2025/3/14 15:50	文件夹
iconengines	2025/3/14 15:50	文件夹
imageformats	2025/3/14 15:50	文件夹
pictures	2025/3/14 16:58	文件夹
platforms	2025/3/14 15:50	文件夹
profilePath	2025/3/14 15:50	文件夹
styles	2025/3/14 15:50	文件夹
tmp	2025/5/27 17:12	文件夹
tools	2025/3/27 17:53	文件夹
translations	2025/3/14 15:50	文件夹
update	2025/1/17 13:32	文件夹
WriteIn_Log	2025/4/28 19:44	文件夹

Figure 1-12

1.2.3 Camera Firmware Reading and Update

A.Ensure All Cables Are Connected

To upgrade the camera firmware, you need to ensure the camera, USB data cable, and Type C cable (Camera Update Cable) are properly connected, as shown in Figure 1-2.

B. Select ISP Model and Load Firmware

Click "Tools" -> "Camera Firmware Upgrade" in the top menu. According to the camera sensor or ISP model that needs to be upgraded, select the correct model. For example, if the camera model to be read this time is SG3-ISX031C-GMSL2F-H190XA, you can enter "031" in the fuzzy search box to quickly locate the specified model. After selecting the model, click "Read Version Number" to read it.

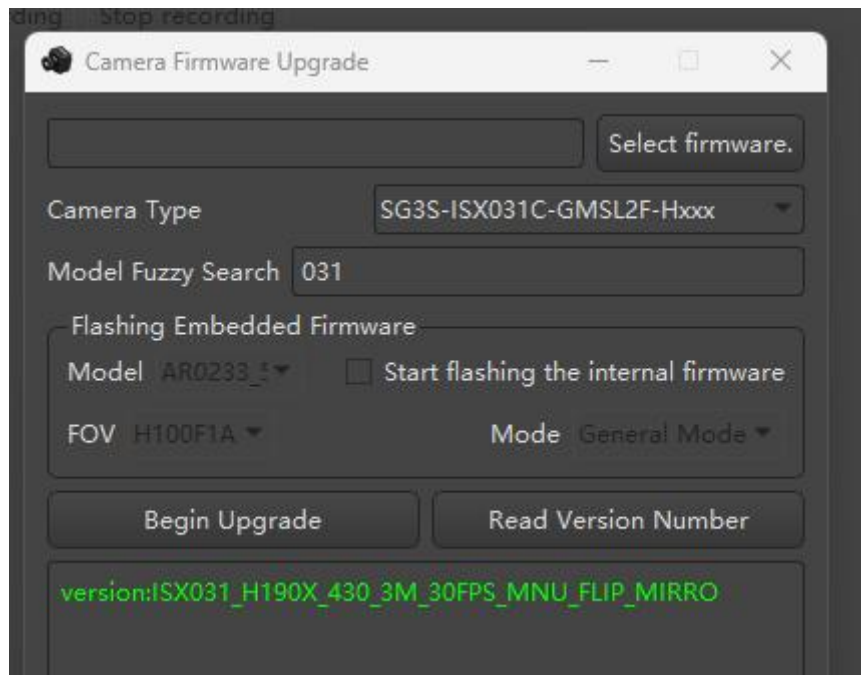


Figure 1-13

C. Firmware Update

- a. In the camera type selection, choose the module model. You can quickly locate it through fuzzy search.
- b. Click the "Select Firmware" button and choose the firmware to be updated in the folder.
- c. Click the left mouse button on the "Start Upgrade" button.

For specific operations, refer to Figure 1-14. After the firmware upgrade is completed, click "Read Version Number". If the version has been updated to the new version, the firmware upgrade is successful, as shown in Figure 1-15

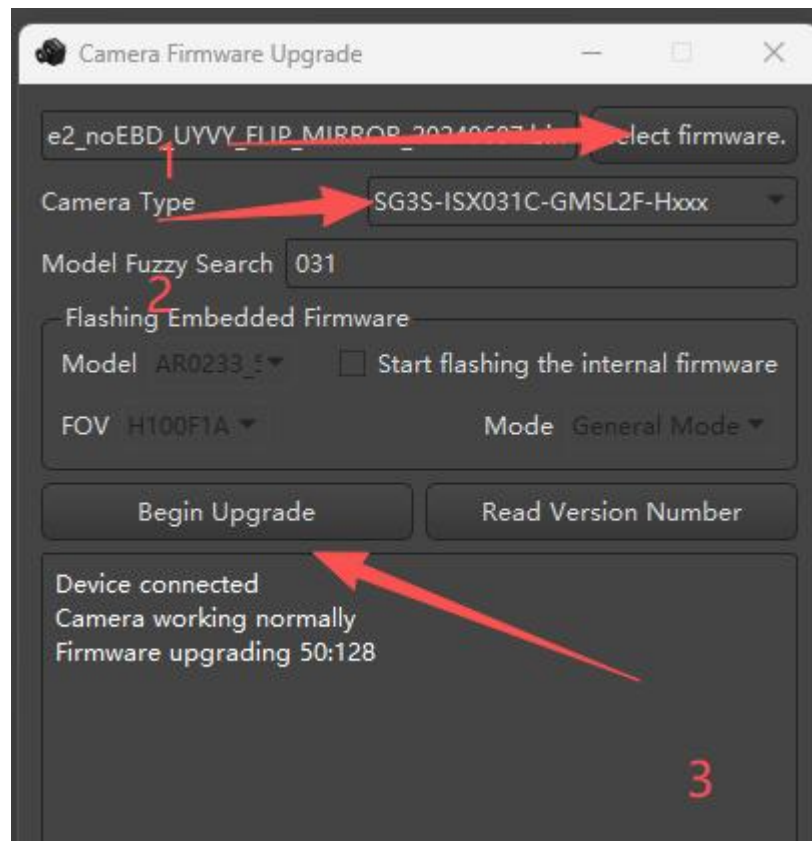


Figure 1-14

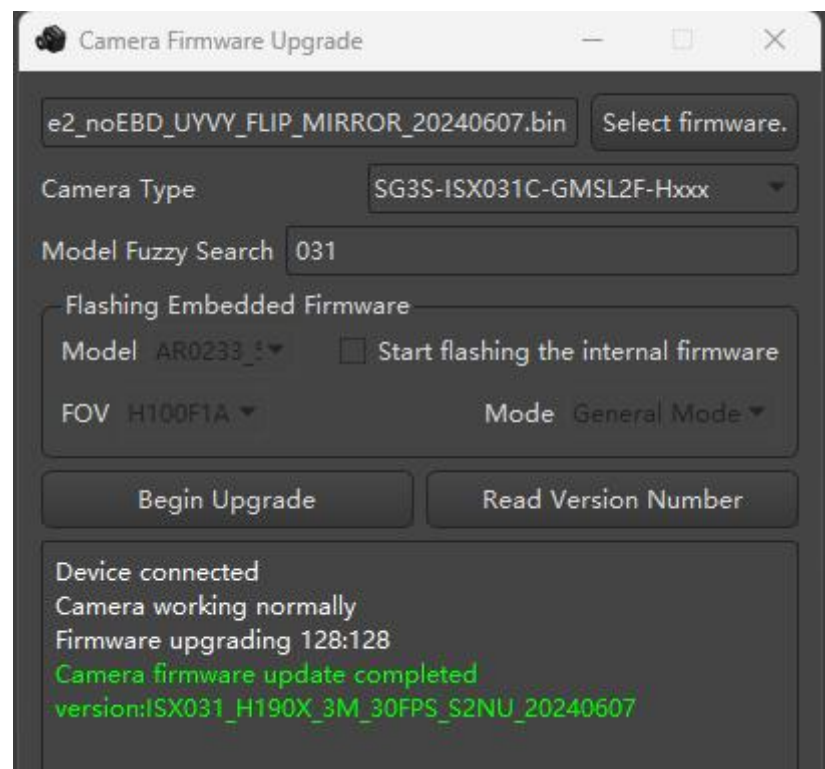


Figure 1-15

D.Switching Modes

Some firmwares are built into the software package, and you can select the specified firmware according to specific usage scenarios. As shown in Figure 1-16, when the module is SG8-AR0820C-G300-H120YA, select the corresponding model, angle and mode, then click "Burn" to proceed.

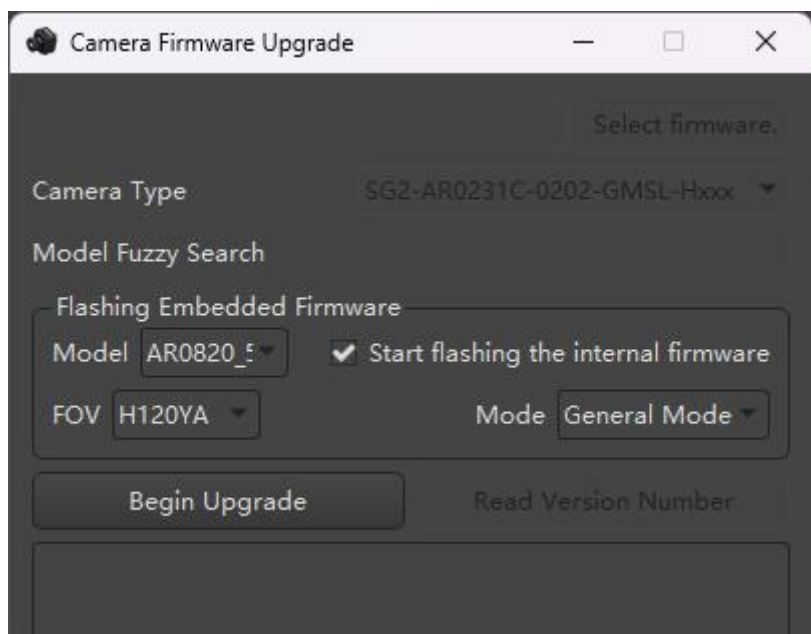


Figure 1-16

1.2.4 Read and Write Registers

In the device information and control interface, the following steps are based on the SG3S-ISX031C-GMSL2F-H190XA example:

A.Fill in the I2C Address

Refer to the product datasheet and hardware configuration page to find the I2C address for the product. For example, the default 8-bit address for the serializer 96717F is 0x80. Enter this address into the I2C address box. The address should be the 8-bit I2C address.

B. Select the Register Width

Select the appropriate register width according to your needs. Currently, both 8-bit and 16-bit widths are supported. For the 96717F, the register width is 16 bits, which can be verified in the product datasheet or hardware configuration page.

C. Fill in the Register Address

Enter the address of the register you want to read or write to. For example, the register address for the trigger signal is 0x02D3, so enter "02D3" in the field.◦

Determine the Read/Write Order: In the read/write order dropdown menu, select the required byte order. You can choose either MSB (Most Significant Byte) first or LSB (Least Significant Byte) first. For this example, select MSB (Most Significant Byte).

D. Select the Register Value Width

Choose 8-bit or 16-bit for the register value width. For the 96717F, the register value width is 8 bits.

E. Input and Operate the Register Value

In the register value field, input the value to be read or written. For example, writing 0x10 to register 0x02D3 will configure GPIO 7 of the 96717F as an output and set it high.

F. Read

Click the "Read" button. If the operation is successful, the value will be displayed as shown in Figure 1-17, with the green text "<read successful> 80, 02D3, 83" indicating a successful read. If the read fails, the text will appear in red as "<read fail> 80, 02D3, err."

G. Write

Click the "Write" button to send the value to the register, as shown in Figure 17. If the write is successful, the text will show in green as "<write successful> 80,02D3, 10." If the write fails, the text will appear in red as "<write fail> 80, 02D3, 10."

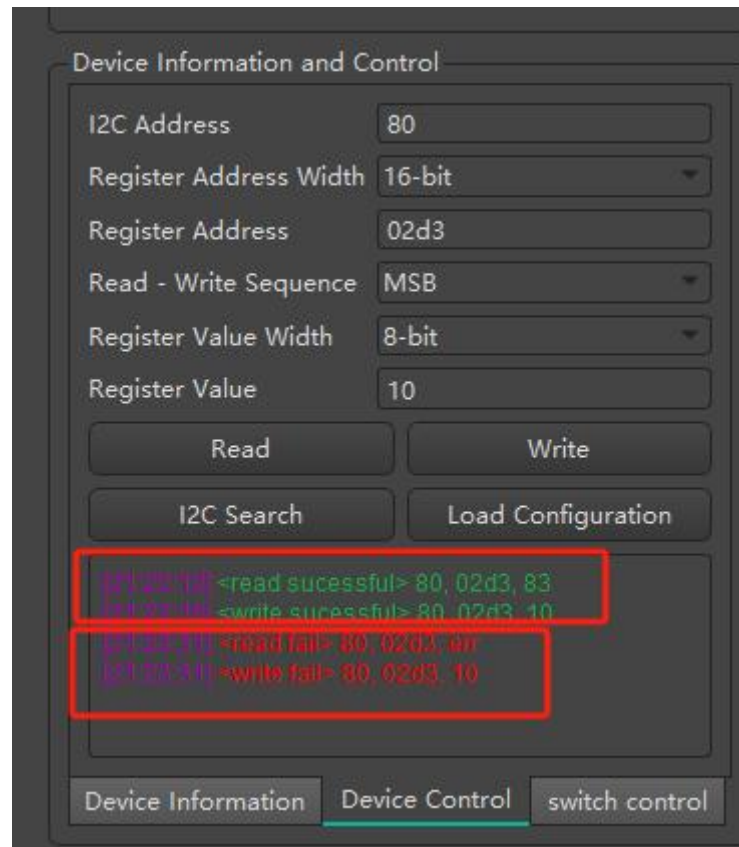


Figure 1-17

1.2.5 External Trigger

Click "Tools" -> "Trigger Signal" in the menu bar at the upper left corner of the main page to open the trigger signal control window, as shown in Figure 1-18. The module is set to external trigger by default. Load the slave configuration built into the software, and the external trigger signal can be sent to the P2 port of the trigger interface, which will then transmit the trigger signal to the module. The trigger interface is shown in Figure 1-19. The external trigger input high level supports either 1.8V or 3.3V.

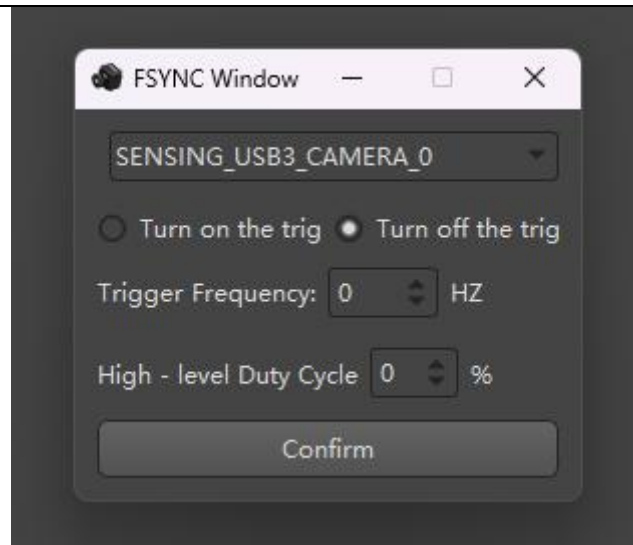


Figure 1-18



Figure 1-19

1.2.6 Current Reading

Connect the camera to PANDA. The current power consumption of the camera can be estimated by reading the current. As shown in Figure 1-20, open the software and select "Tools" -> "Channel Control" in the upper left menu to open the current reading interface. After opening the interface (as shown in Figure 1-21), click "Get Current" to display the current on the corresponding port. If no external power supply is used and the USB port's internal power supply is adopted, the POC voltage is approximately 11.8V. According to the current read as shown in Figure 1-21, the calculated power is approximately 11.8 *

0.078 = 0.920W.

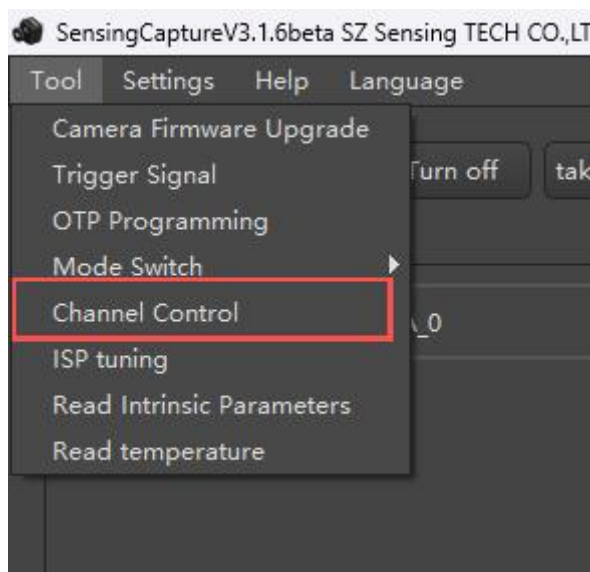


Figure 1-20

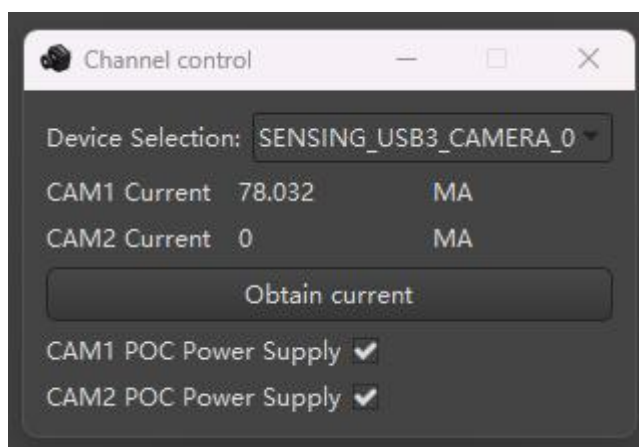


Figure 1-21

1.2.7 Camera Intrinsic Parameter Reading

If the camera has undergone internal parameter calibration, you can read the camera's internal parameters through the "Read Internal Parameters" function of the upper computer. As shown in Figure 1-22, on the main interface, select "Tools" -> "Read Internal Parameters" in the menu bar at the upper left corner. In the opened interface, click "Read Internal Parameters" to start reading. Figure 1-23 shows the read internal parameters of a binocular camera.

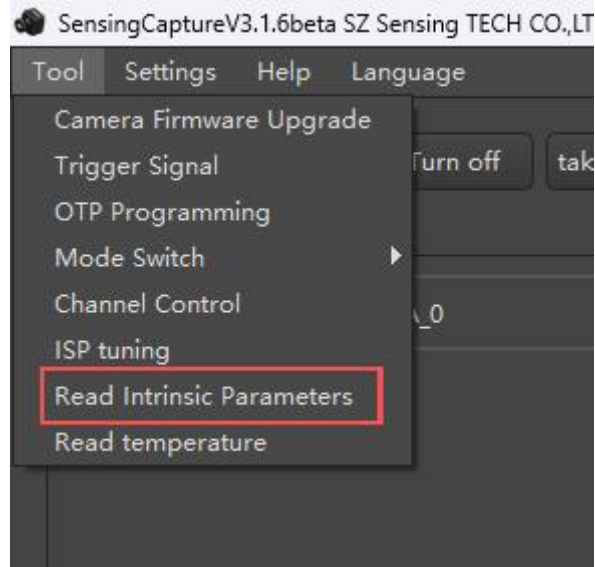


Figure 1-22

If you need to save the internal parameter data, click "Export Parameters" to save the read internal parameter data as a TXT file.

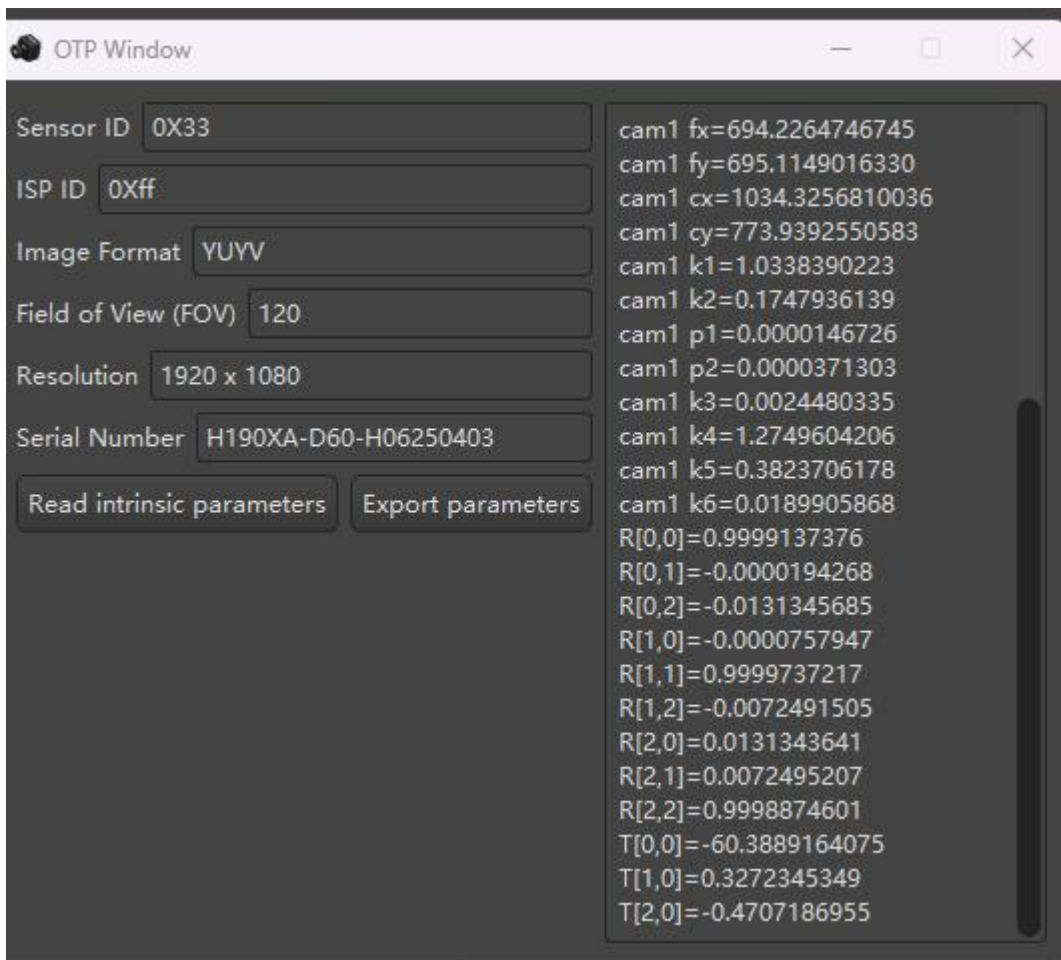


Figure 1-23

1.2.8 PANDA Firmware Update

For the first programming, you need to install the driver, which is located in 3rdparty/DriverInstallerCyUSB3 under the program directory.

Step 1: As shown in Figure 1-24, there is a button in the framed area. Press and hold this button (boot button), then connect the USB3.0 interface to the computer (make sure the button is pressed when power is applied).



Figure 1-24

Step 2: Open the host computer software. In the upper left corner, go to Settings and select "Device Firmware Upgrade", as shown in the framed area in Figure 1-25.

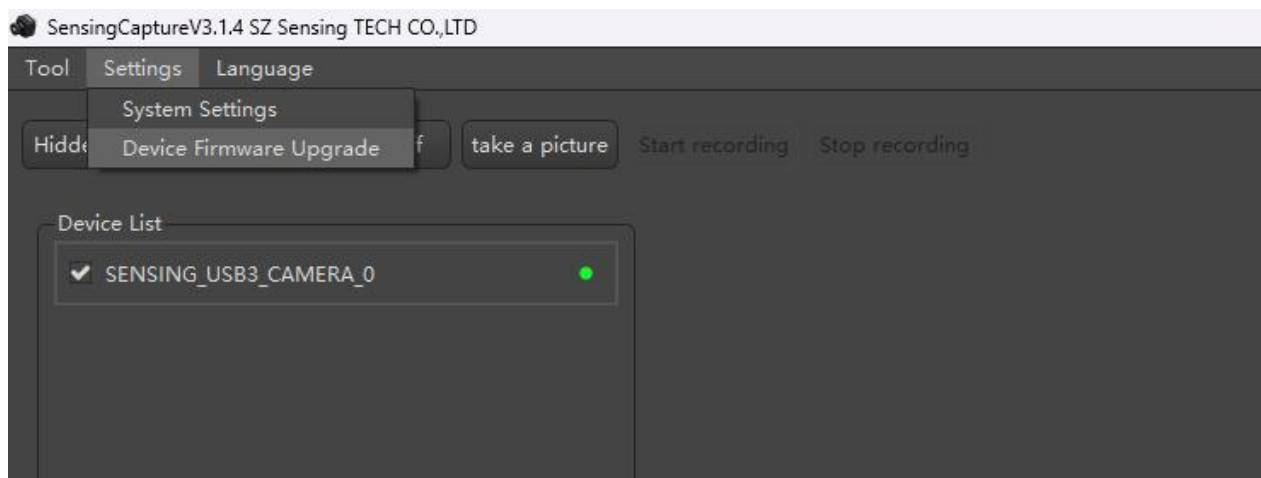


Figure 1-25

Step 3: If the device is recognized as being in upgrade mode, "Cypress FX3 USB BootLoader Device" will appear in the left list of the pop-up window. If it does not appear,

you need to restart the process from Step 1.

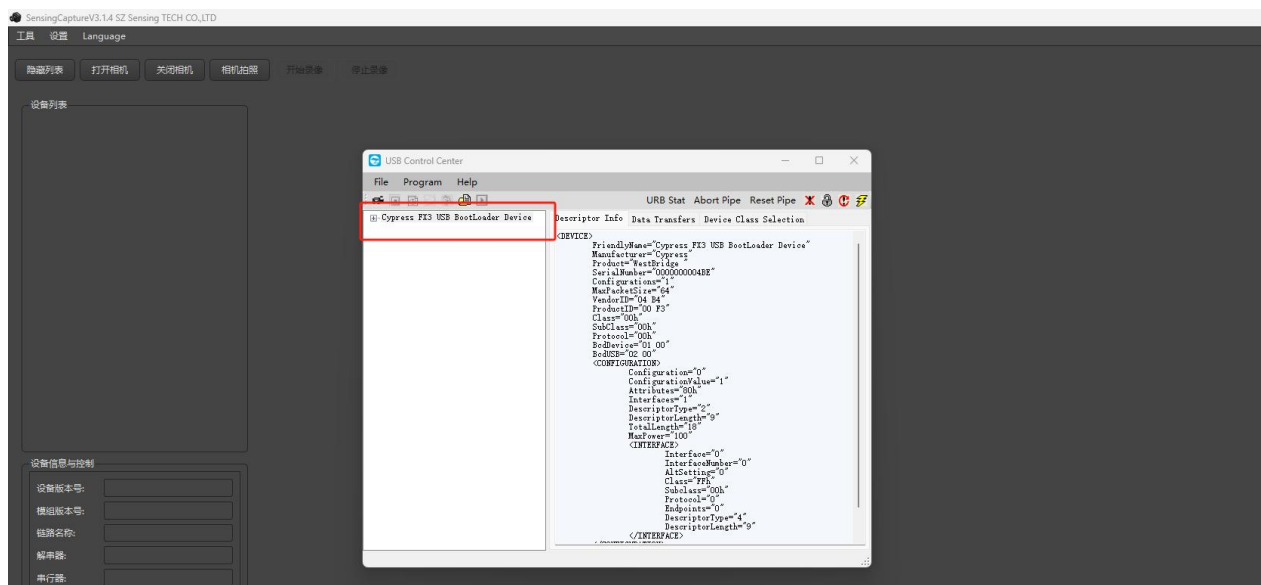


Figure 1-26

Step 4: As shown in Figure 1-27, in the upgrade interface, select "I2C EEPROM". After selection, a pop-up window for path selection will appear. Choose the firmware to be upgraded and click "Open".

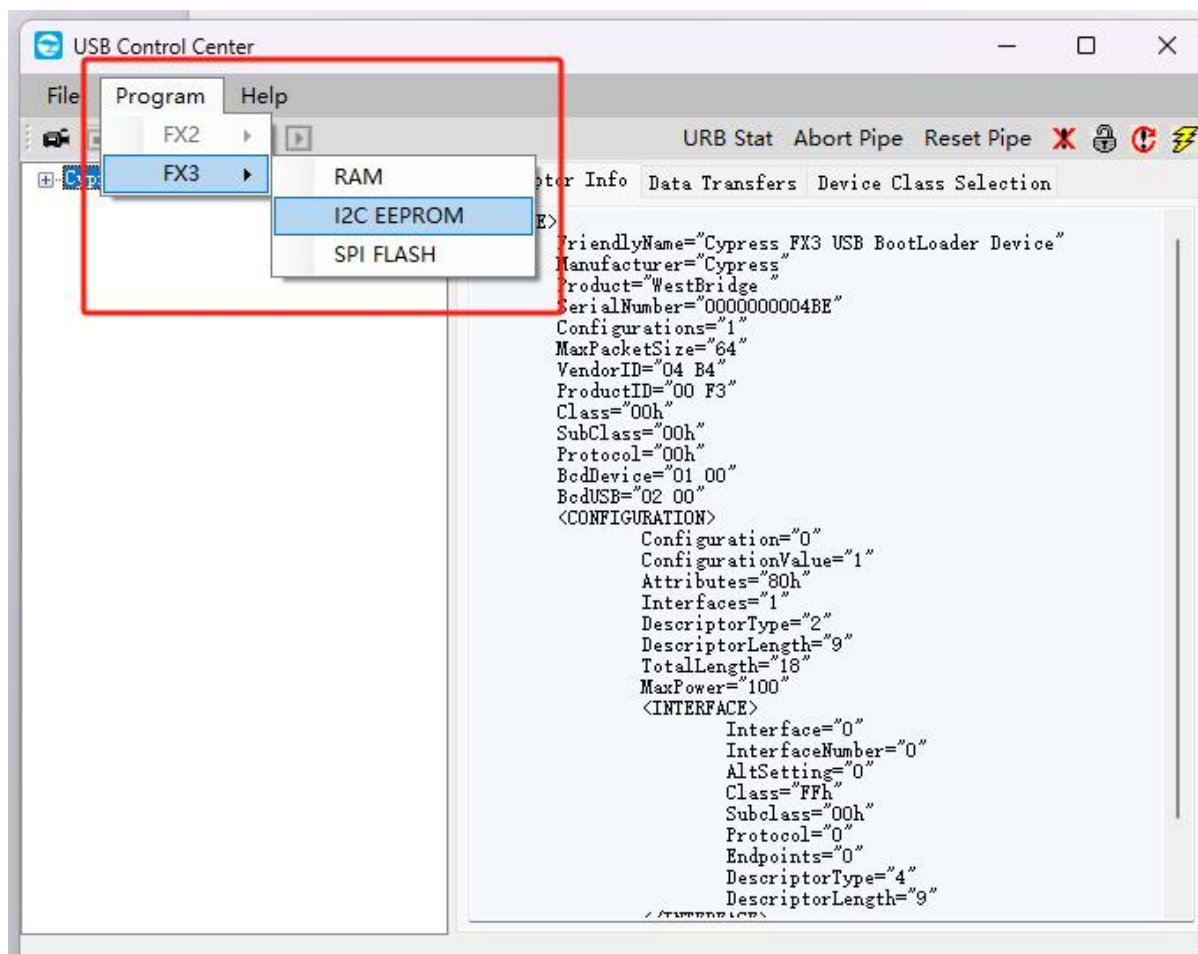


Figure 1-27

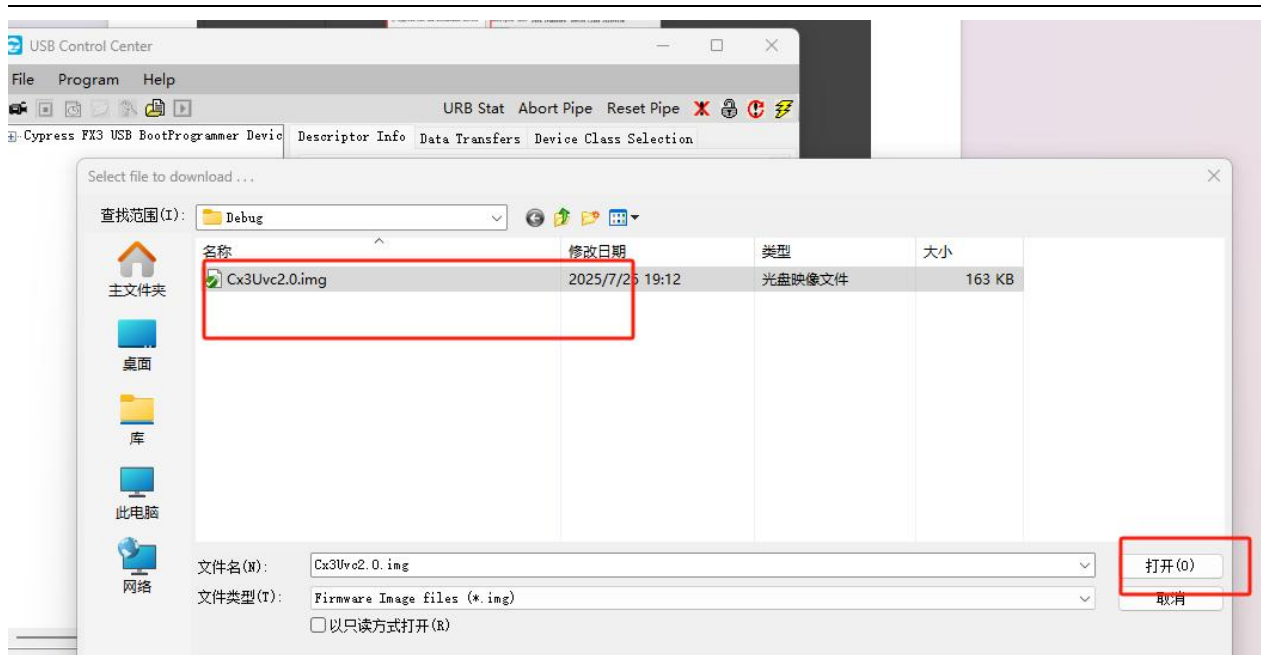


Figure 1-28

Step 5: After selection, "Upgrading" will be displayed in the lower left corner as shown in Figure 1-29. Wait until "Succeeded" is displayed as shown in Figure 1-30, which indicates that the upgrade is successful.

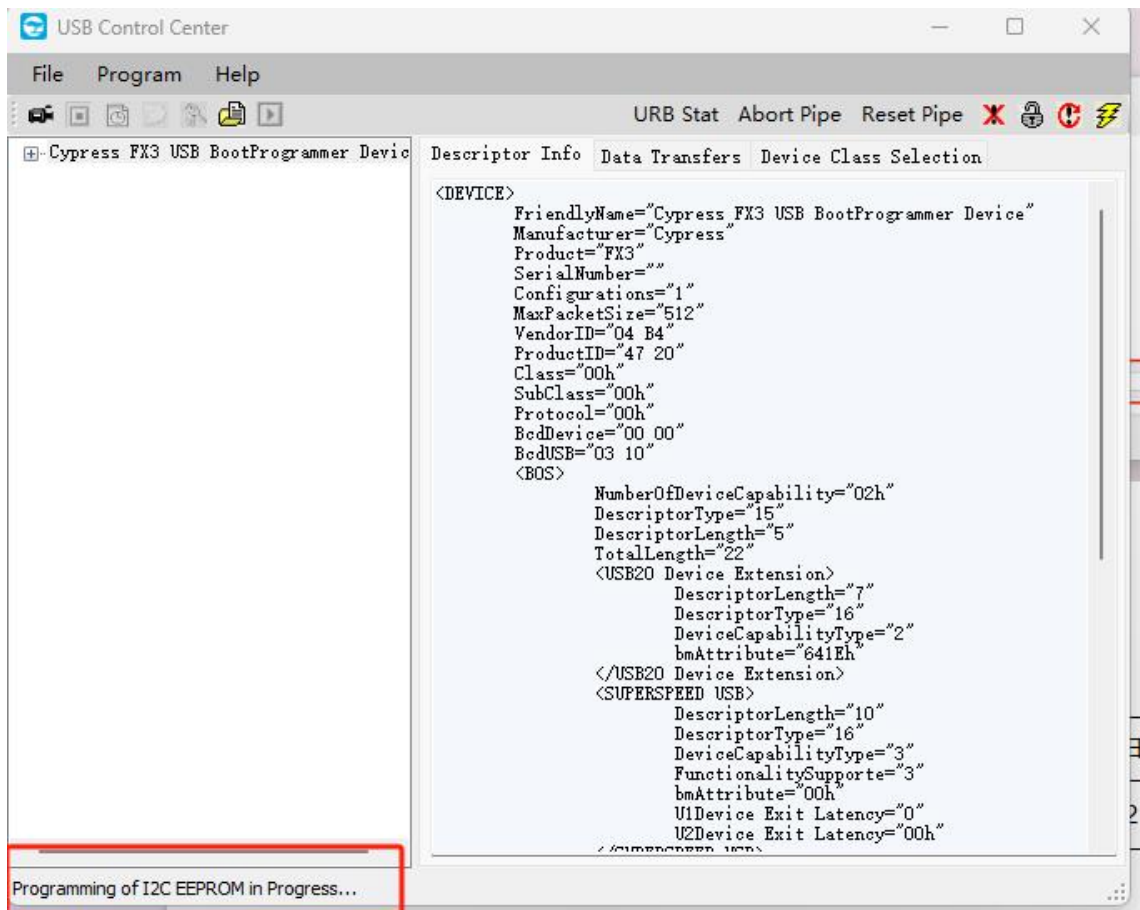


Figure 1-29

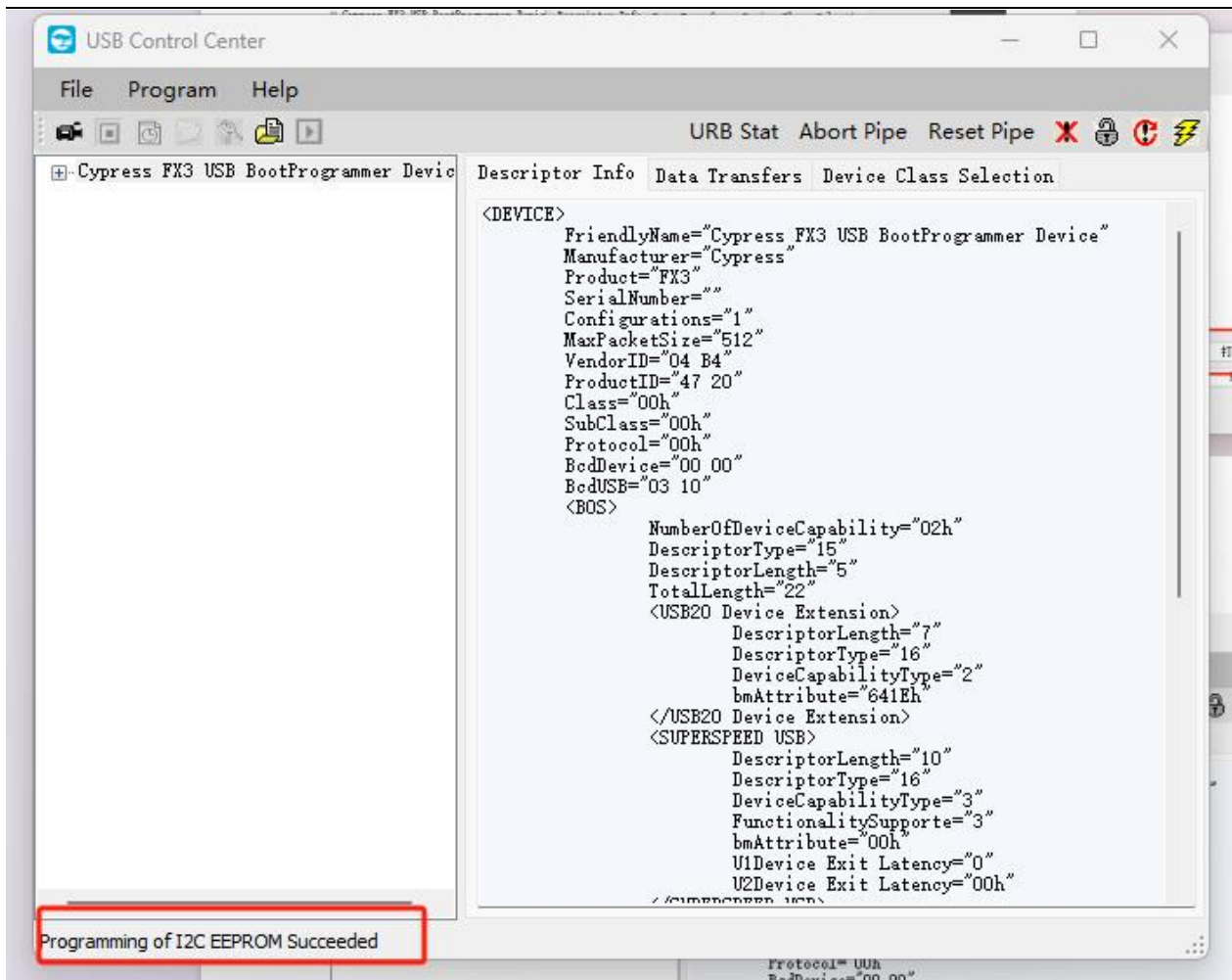


Figure 1-30

1.2.9 ISP Tuning

You can easily adjust the image effects through the upper computer's UI interface. The adjusted effects will not be saved in the firmware. When the module is restarted, it will return to the firmware's default effect state. However, the adjusted parameters can be imported and exported. As shown in Figure 1-31, on the main interface, select "Tools" -> "ISP Tuning" in the menu bar at the upper left corner to open the effect adjustment interface.

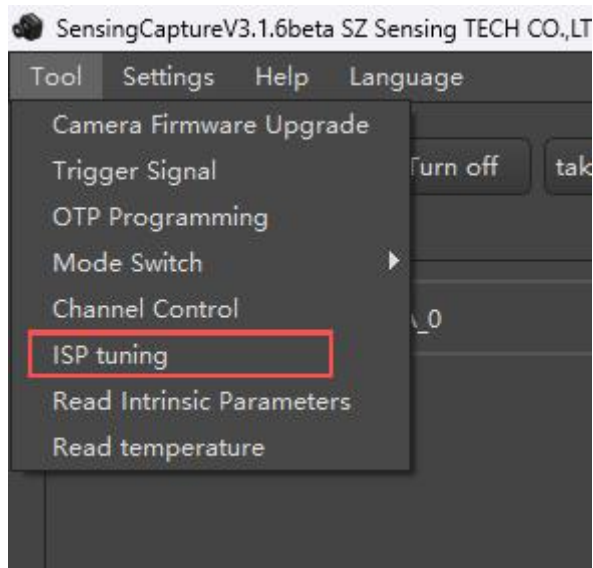


Figure 1-31

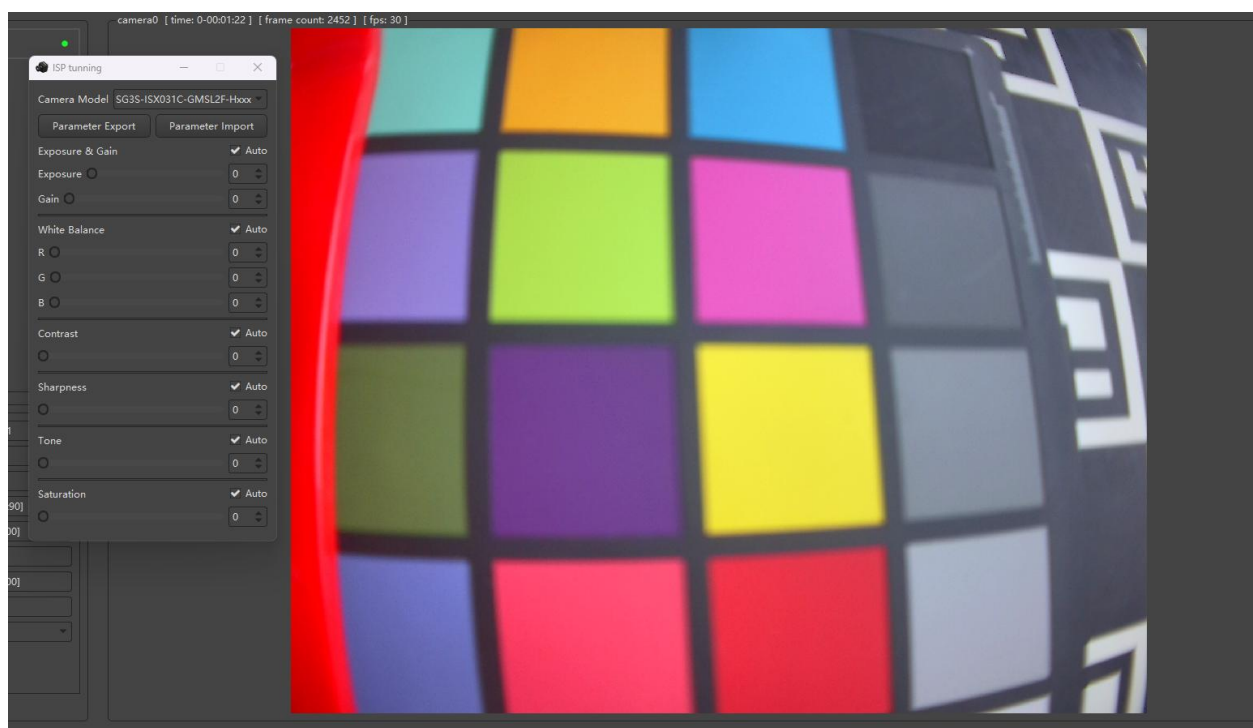


Figure 1-32

After opening the interface, select the corresponding model. For example, if using SG3-ISX031C-G2F, the default effect after opening is as shown in Figure 1-32.

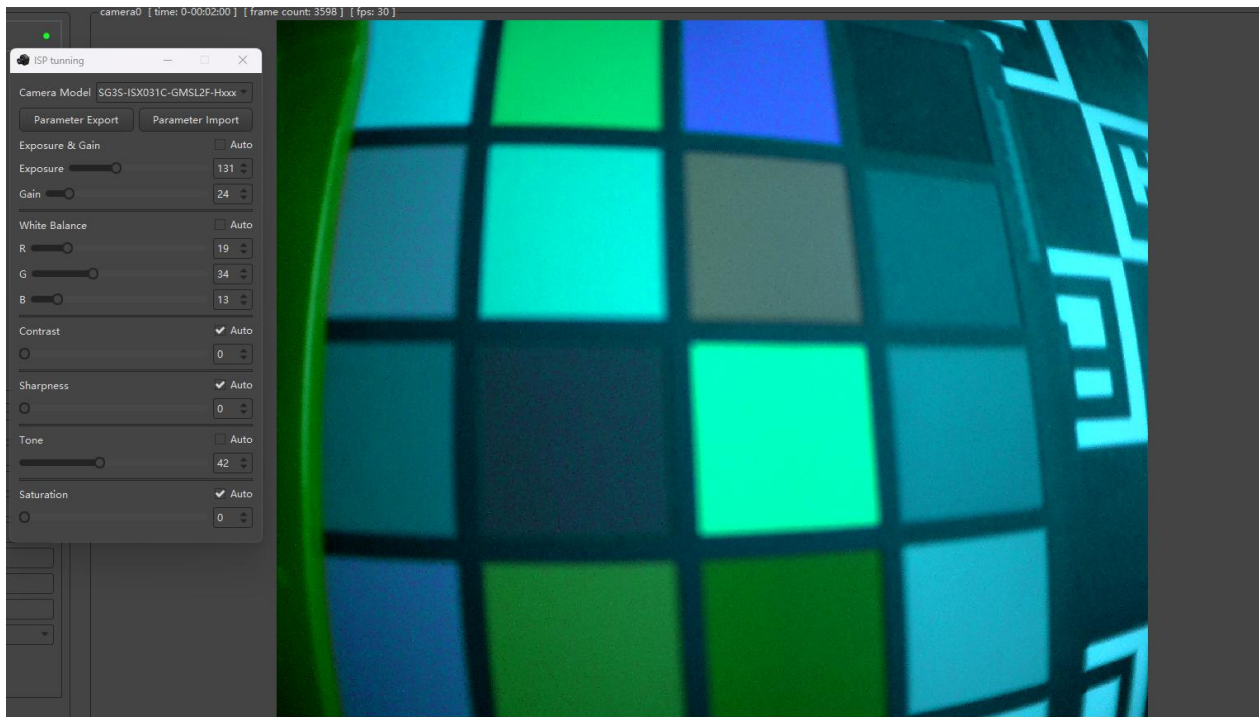


Figure 1-33

After adjusting parameters such as exposure, gain, and white balance through the interface (as shown in Figure 1-33), click the "Export Parameters" button to save the current parameters as a configuration file. When reopening, click "Import Parameters" and select the corresponding configuration file to apply them.

1.2.10 Read lens temperature

Some modules (RedFox-D3Gx, CMS) are equipped with a sensor in the lens, through which the current temperature inside the lens can be obtained. On the main interface, click "Tools" in the upper left corner and then select "Read Temperature" to open the temperature reading interface as shown in Figure 1-34. After opening, click the "Get Temperature" button to read the temperature.

Note: This temperature refers to the temperature at the lens sensor, not the temperature on the lens surface.

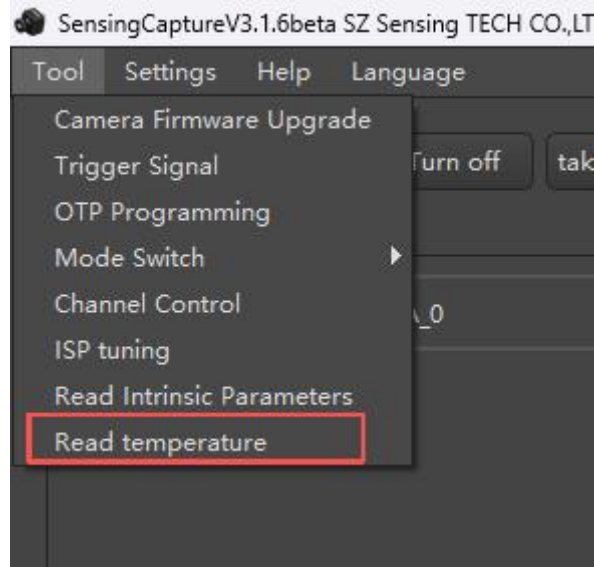


Figure 1-34

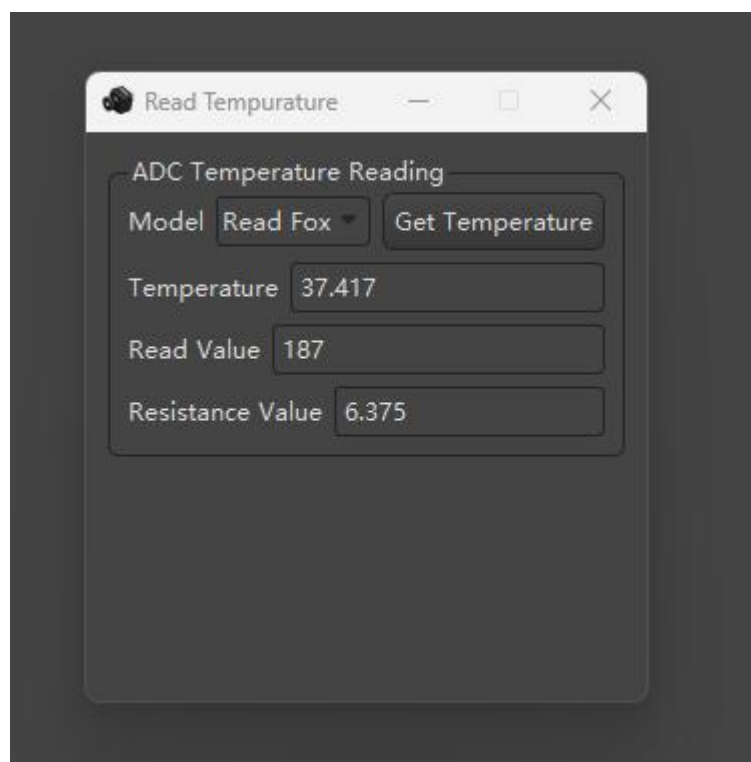


Figure 1-35

2. Device User Guide for Linux System

2.1 Software Download

Supported Systems (X86): ubuntu18.04 , ubuntu20.04 , ubuntu22.04

Download Commands:

git clone -b Linux https://github.com/SENSING-Technology/SensingCaptureV3.git

2.2 Usage Instructions

2.2.1 Launch the Host Computer Software

Use the command `lsusb | grep Cypress` to verify.

If the following device information appears: ID 04b4:00c3 Cypress Semiconductor Corp., the system has recognized the device.

```
putureV3_linux-Desktop Qt 5 12 12 GCC 64bit-Debug$ lsusb | grep Cypress
Bus 002 Device 003: ID 04b4:00c3 Cypress Semiconductor Corp.
```

Install Some Dependencies

```
sudo apt install libudev-dev
```

```
sudo apt-get install i2c-tools
```

Navigate to the Software Folder Path, right-click to open it in the terminal. Enter the command `sudo ./sensingCaputureV3_linux`

Permission needs to be added for the first activation:

```
chmod 777 ./sensingCaputureV3_linux
```

The opened interface is as follows

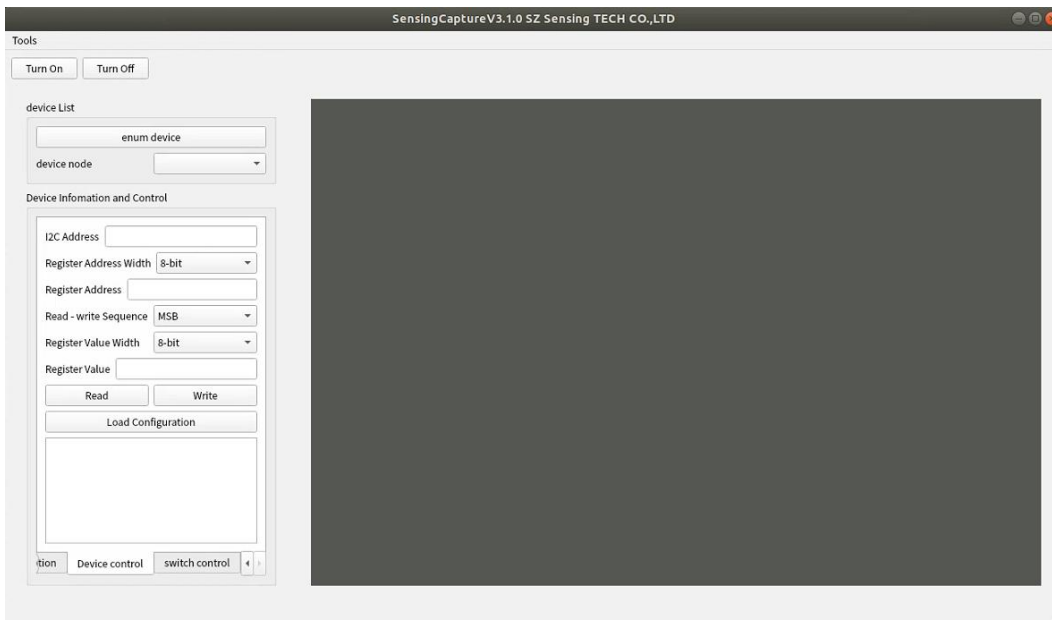


Figure 2-1

2.2.2 Load Configuration to Activate the Camera

Camera activation steps:

A. On the Device Control interface, click the "Load Configuration" button.

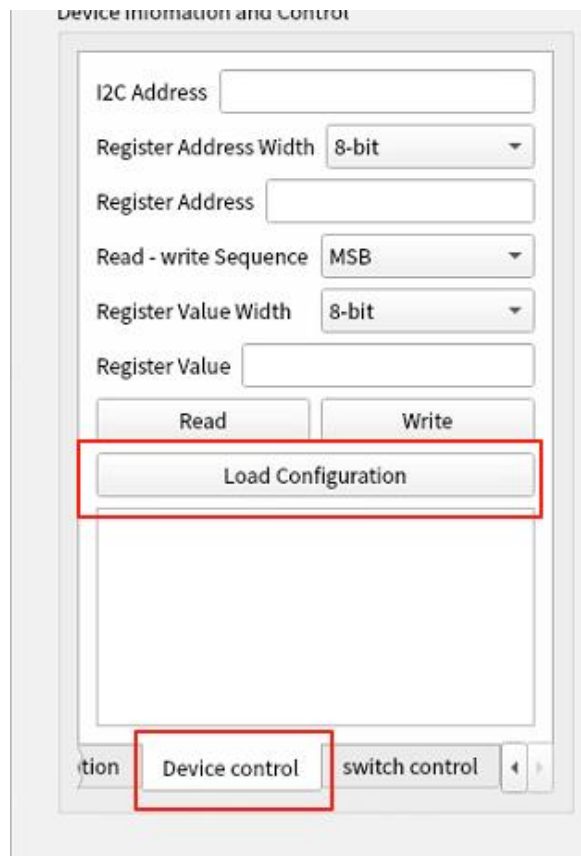


Figure 2-2

B. In the pop-up window, first enter the key information of the model in the input box after "fuzzy search", then open the drop-down box to select the corresponding configuration file. After selecting the configuration file, click "confirm" to start loading the configuration file.

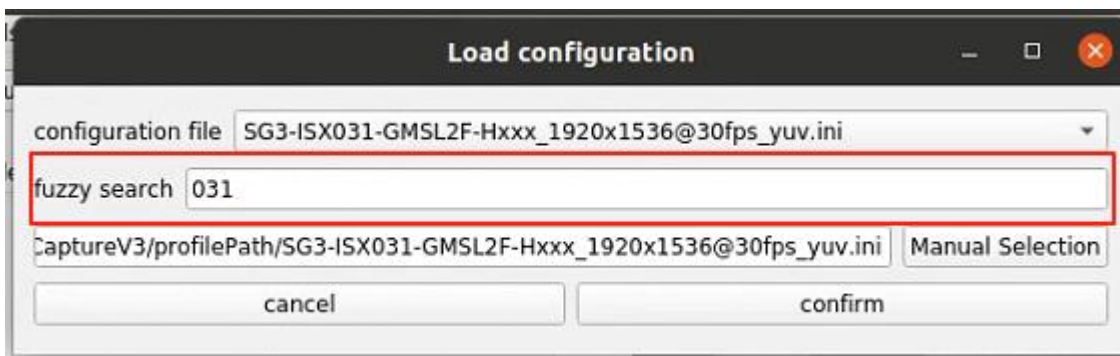


Figure 2-3

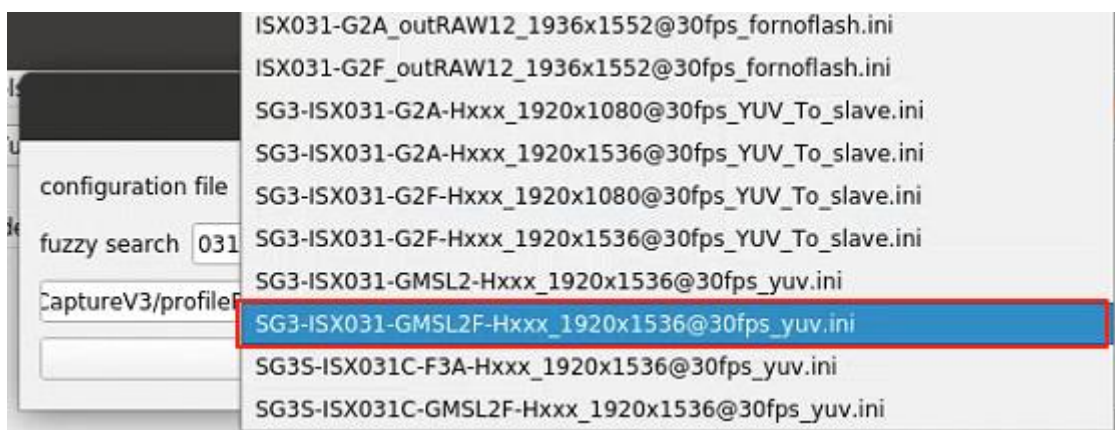


Figure 2-4

If "reconnect" appears, the configuration file has been loaded successfully.

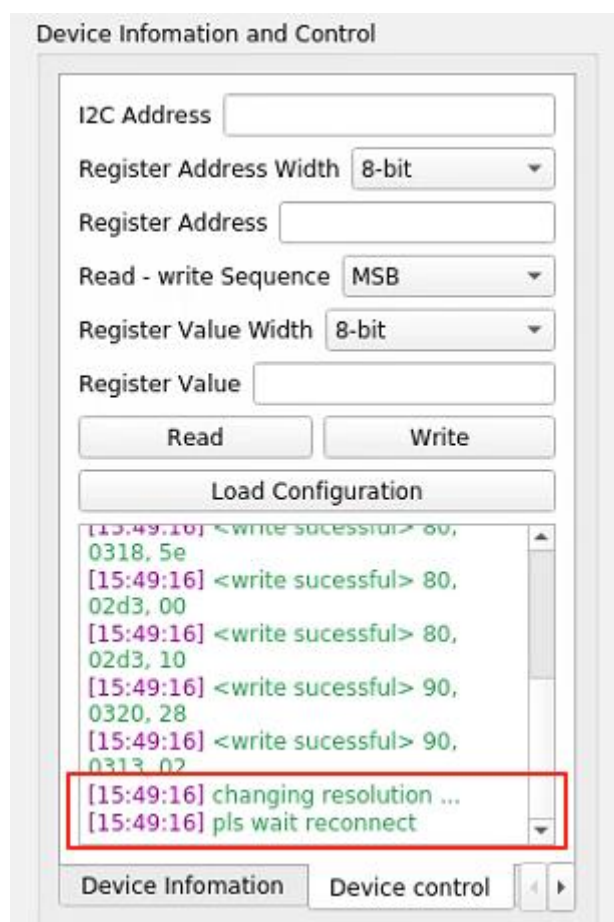


Figure 2-5

Enumerate Device Nodes: When Panda is connected to a Linux system, the system will recognize and generate 2 new video nodes. Select the first one among these two nodes. The video0 node is opened by default when the camera is turned on. If there are multiple camera devices, you need to manually select the corresponding node.

Use the command `v4l2-ctl -d /dev/videox --info` (x needs to be replaced with the system node).

If the card type shown in the figure is: SENSING_USB3_CAMERA, then this node is the device node .of Panda

```
guo@guo-TianYi510Pro-14ACN:~$ v4l2-ctl -d /dev/video0 --info
Driver Info:
  Driver name      : uvcvideo
  Card type        : SENSING USB3 CAMERA
  Bus info         : usb-0000:01:00.0-2
  Driver version   : 5.15.178
  Capabilities     : 0x84a00001
                    Video Capture
                    Metadata Capture
                    Streaming
                    Extended Pix Format
                    Device Capabilities
  Device Caps      : 0x04200001
                    Video Capture
                    Streaming
                    Extended Pix Format
guo@guo-TianYi510Pro-14ACN:~$
```

Figure 2-6

Select the corresponding node based on the read node information.

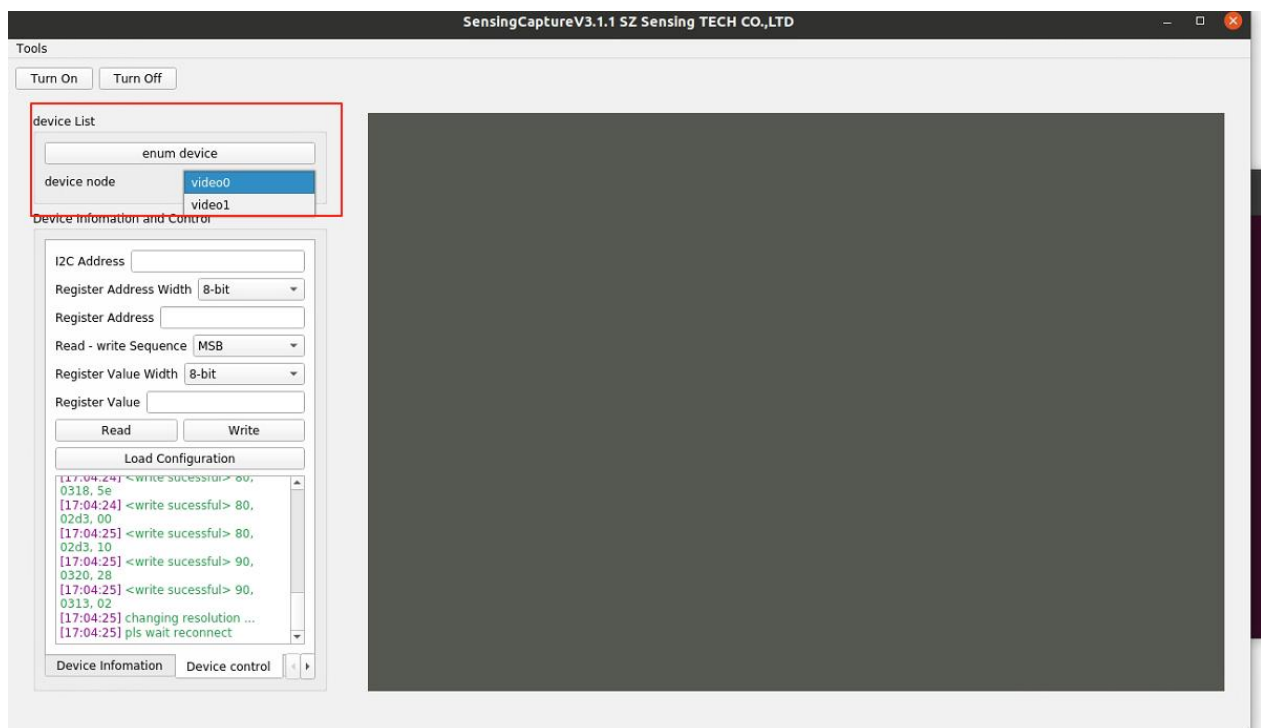


Figure 2-7

Click the "Turn On" button to activate the camera.

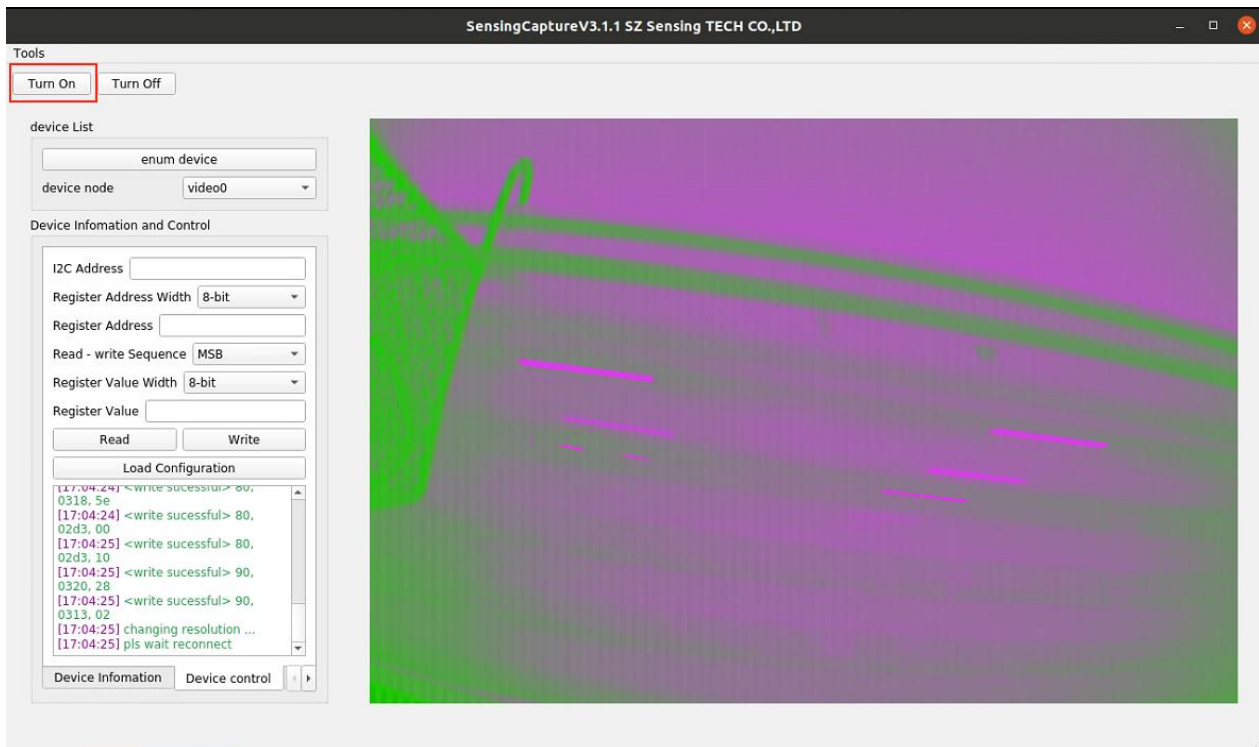


Figure 2-8

The system parses images in YUYV format by default. If the image is in UYVY format, format conversion is required. On the Device Information interface, select UYVY from the drop-down box of Image Parsing Format. As shown in the figure, the format has been successfully converted.

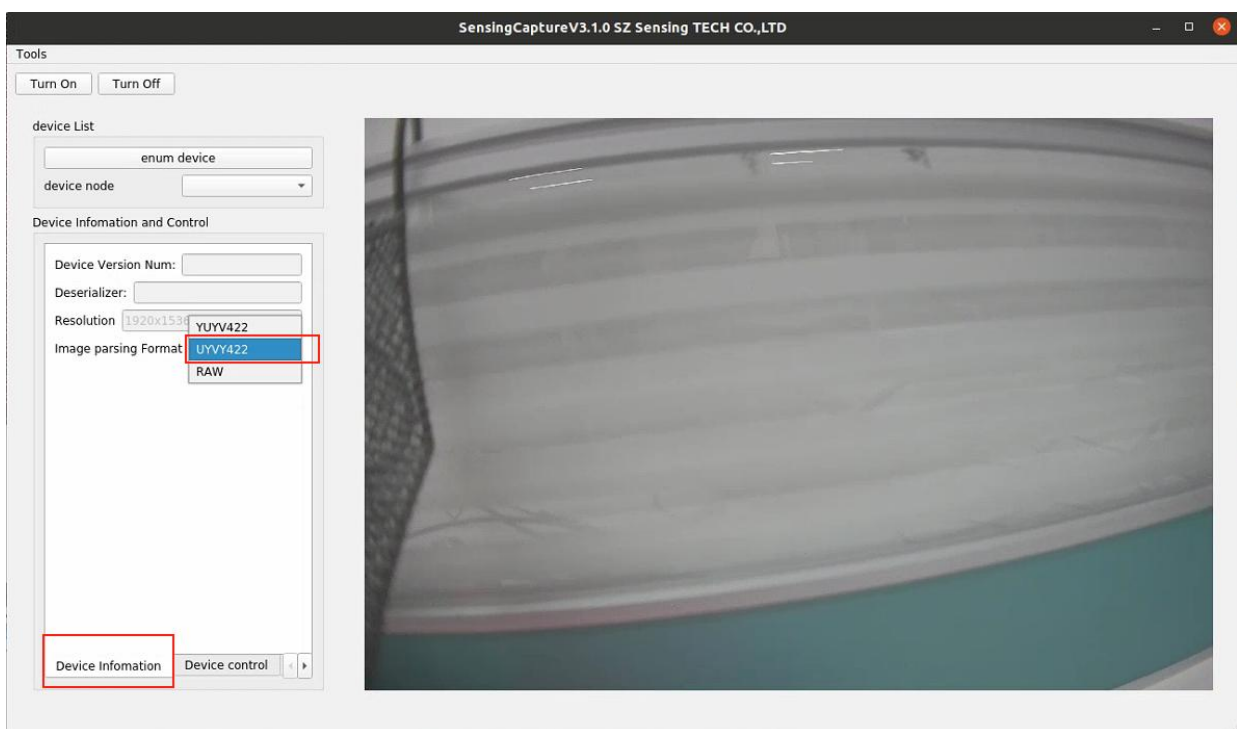


Figure 2-9

2.2.3 Read and Write Registers

The host computer supports reading and writing registers. The method is as follows:

Enter the I2C address in the input box after "I2C Address". For example, to read the working mode of MAX9296 as shown in the figure, the I2C address of MAX9296 is 0x90, so enter 90 in the input box.

Select the register address width: Choose the corresponding register width as needed. Currently, 8bit and 16bit are supported. The register width of MAX9296 is 16bit, which can be found in the product specification, hardware configuration page, or MAX9296's DATASHEET.

Enter the register address to be read or written in "Register Address". For example, if you need to read the working mode of MAX9296, whose corresponding register address is 0x1, enter 1 in the input box after "Register Address".

Select "Read-write Sequence". Choose whether the high byte comes first or the low byte comes first. This option can be ignored if writing 8-bit data. If writing 16-bit data and the low byte data needs to come first, select LSB; in other cases, keep it as MSB.

Select the width of the value to be written: The value width can be 8-bit or 16-bit. Choose according to the manual of the device to be read or written. For example, in this example, if you need to read and write 8-bit data, select 8-bit.

Enter the value to be written: If you need to write a value, enter the value to be written in the input box after "Register Value"; if you are reading a register, the read value will be displayed in this input box after clicking "Read".

After confirming the information for writing or reading, click the "Read" button to perform the read operation, or click the "Write" button to perform the write operation.

As shown in the figure, if the data is read successfully, "<read successful>" will be displayed in the display box.

If the data is written successfully, "<write successful>" will be displayed in the display box.

If there is an error in reading data, "<read fail>" will be displayed in the display box.

If there is an error in writing data, "<write fail>" will be displayed in the display box.

Note: Reading and writing registers during image output will interrupt the image

output. You need to re-perform the camera activation steps after completing the operation.

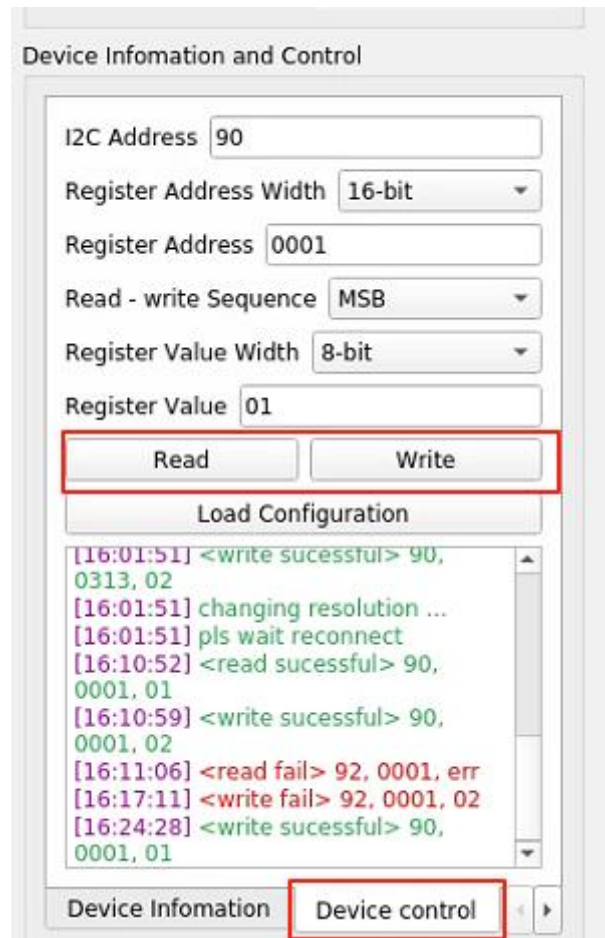


Figure 2-10

2.2.4 Switch Control

On the Switch Control interface, you can read the current and control the camera's power switch.

When the camera is connected to the Panda, you can estimate the camera's current power consumption by reading the current. Click the Obtain Current button to get the current value and view the current of the module. As shown in the figure, the read current is 104.32 mA (note: "MA" is corrected to lowercase "mA"—the standard unit symbol for milliamperes in technical contexts).

The checkbox next to Channel 1 is used to control the POC (Power over Coax) power supply of the CAM1 port. If you uncheck it, the POC power supply for CAM1 will be disconnected.

The checkbox next to Channel 2 is used to control the POC power supply of the CAM2 port. If you uncheck it, the POC power supply for CAM2 will be disconnected.

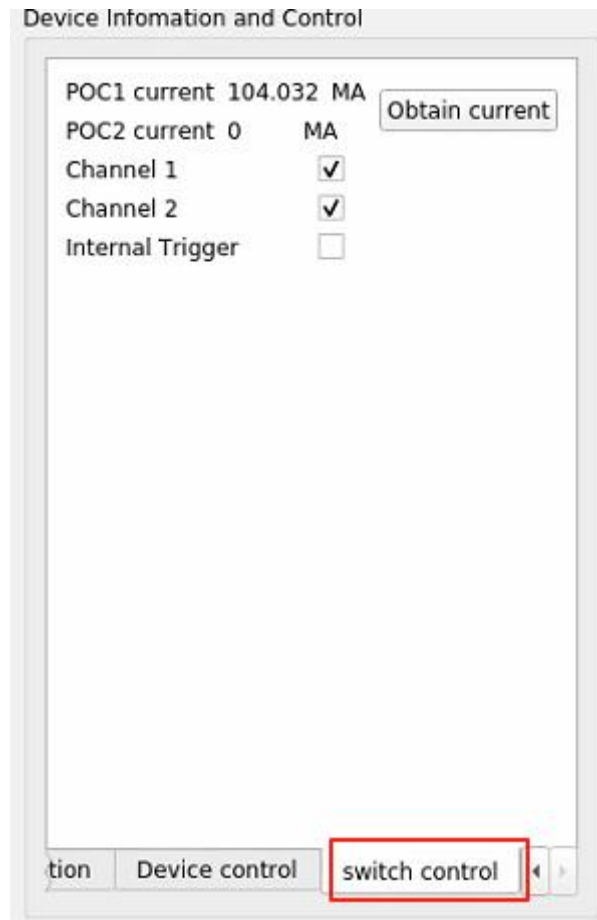


Figure 2-11

2.3 Use Scripts in a Linux System

In the default firmware of the PANDA , only modules with a resolution of 1920*1080 can be lit up. To light up modules with other resolutions, you need to load the configuration first. The configuration file is the same as that used in the Windows system.。

First, obtain the program for loading configurations in Linux.

Download link: <https://github.com/SENSING-Technology/PandaCtrl>

As shown in Figure 29, the program package should contain the "PandaCtrl" folder and the source code files of libusb. The program files come with a libusb library compiled for the Ubuntu 20.04 system by default. If you need to use it on other systems, you need to recompile the libusb library, replace the generated library files, and then compile the program for loading configuration files.

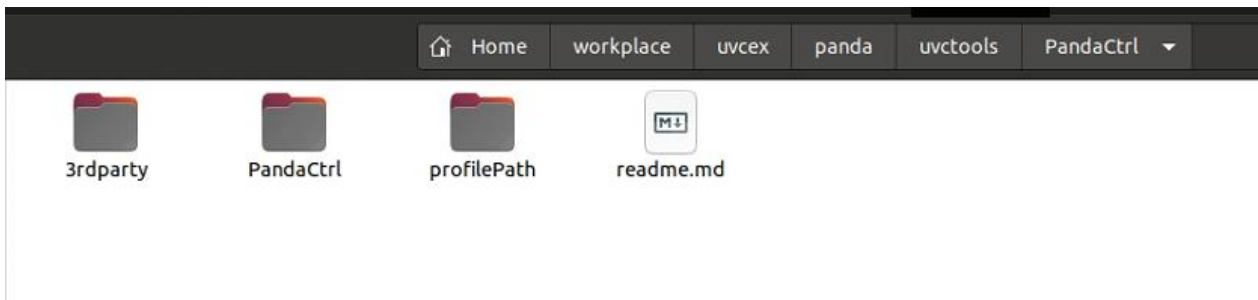


Figure 2-12

Open the PandaCtrl folder and run the makefile inside it (enter make in the command line).

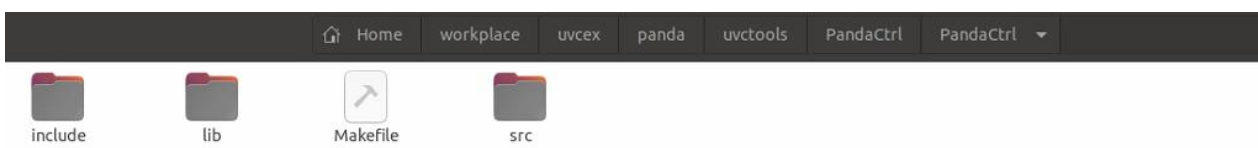


Figure 2-13

After executing the makefile, if the compilation is successful, an executable program named PandaCtrl will be generated.

Note: You need to install the gcc and make tools first.

Some systems require additional dependency libraries to be installed. For example, on Ubuntu 20.04, you need to execute the following commands:

```
sudo apt install libudev-dev
```

```
sudo apt-get install i2c-tools
```

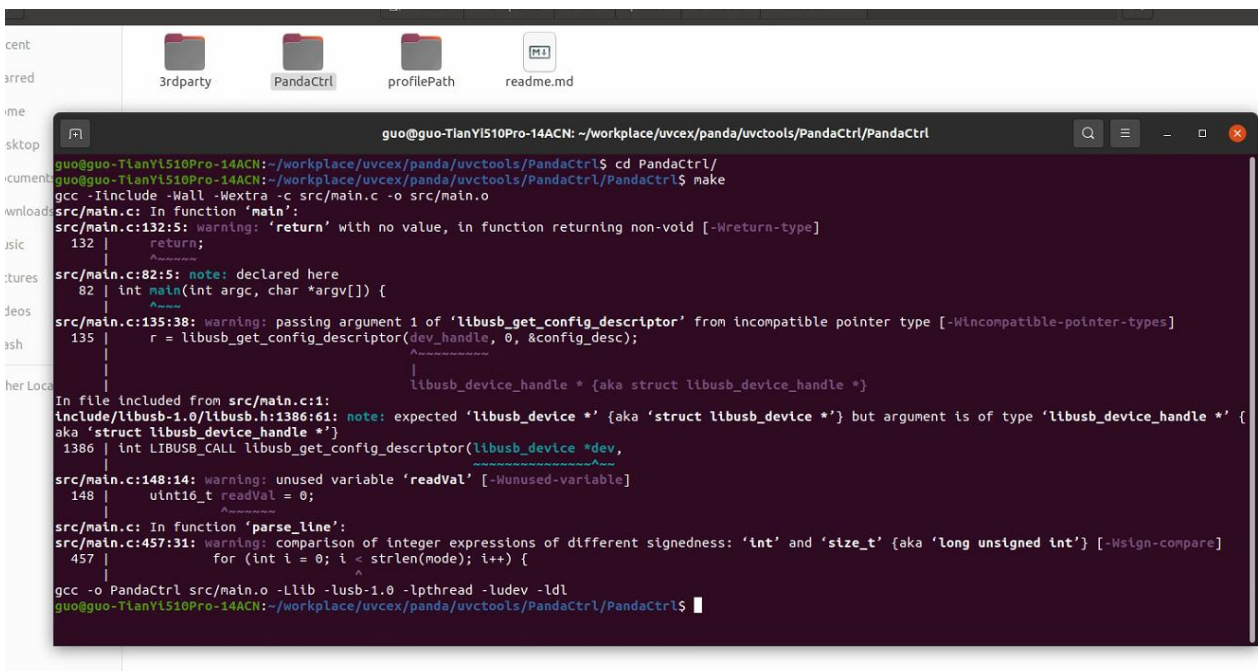


Figure 2-14

The command format for execution is: `sudo ./PandaCtrl` [absolute path of the configuration file]

As shown in Figure 15, after running the command, if you see the prompt "Device opened successfully", it indicates that the configuration file has been loaded normally.

```

guo@guo-TianYi510Pro-14ACN: ~/workplace/uvnex/panda/uvctools/PandaCtrl/PandaCtrl$ make
gcc -Iinclude -Wall -Wextra -c src/main.c -o src/main.o
src/main.c: In function 'main':
src/main.c:132:5: warning: 'return' with no value, in function returning non-void [-Wreturn-type]
132 |     return;
    |     ^~~~~~
src/main.c:82:5: note: declared here
82 | int main(int argc, char *argv[]) {
    | ^~~~~
src/main.c:135:38: warning: passing argument 1 of 'libusb_get_config_descriptor' from incompatible pointer type [-Wincompatible-pointer-types]
135 |     r = libusb_get_config_descriptor(dev_handle, 0, &config_desc);
    |                                     ^~~~~~
    |                                     |
    |                                     libusb_device_handle * {aka struct libusb_device_handle *}
In file included from src/main.c:1:
include/libusb-1.0/libusb.h:1386:61: note: expected 'libusb_device **' {aka 'struct libusb_device **'} but argument is of type 'libusb_device_handle **' {aka 'struct libusb_device_handle **'}
1386 | int LIBUSB_CALL libusb_get_config_descriptor(libusb_device *dev,
    |                                             ^~~~~~
src/main.c:148:14: warning: unused variable 'readVal' [-Wunused-variable]
148 |     uint16_t readVal = 0;
    |
src/main.c: In function 'parse_line':
src/main.c:457:31: warning: comparison of integer expressions of different signedness: 'int' and 'size_t' {aka 'long unsigned int'} [-Wsign-compare]
457 |     if (strlen(line) > 1023) {
    |         ~~~~~^~~~~~
guo@guo-TianYi510Pro-14ACN:~/workplace/uvnex/panda/uvctools/PandaCtrl$ sudo ./PandaCtrl /home/guo/workplace/uvnex/panda/uvctools/PandaCtrl/profilePath/SG35-ISX0316
[sudo] password for guo:
Kernel driver active, detaching...
Device opened successfully
Unable to claim interface
guo@guo-TianYi510Pro-14ACN:~/workplace/uvnex/panda/uvctools/PandaCtrl$

```

Figure 2-15

After the loading is complete, open the system camera. If the camera can be opened normally, it indicates that the configuration is successful. If the camera can display images but the color gamut is incorrect, you can refer to the conversion between UYVY and YUYV formats in the appendix.。

As shown in Figure 2-16, If you need to recompile the libusb library, after compiling it in libusb-1.0.26, replace the PandaCtrl/lib/libusb-1.0.a file with the libusb/.lib/libusb-1.0.a file.

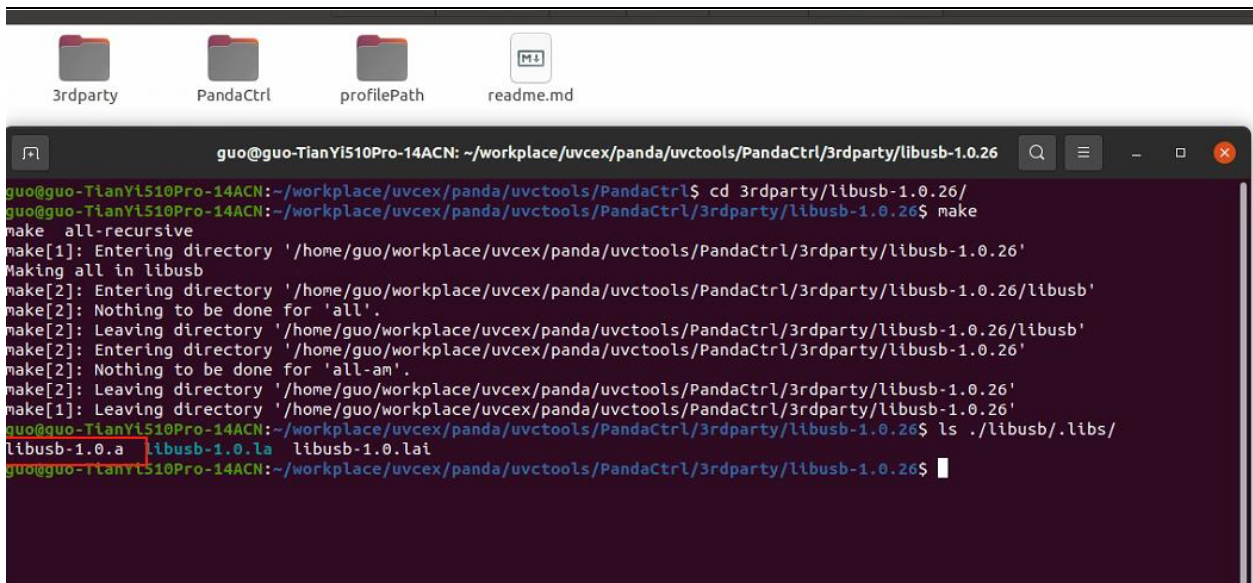


Figure 2-16

3.Version History

Version	Description	Date
Ver 1	First release	2025/08
Ver 2	Improve the content for use on the Linux system.	2025/09
Ver 3	Improve Usage Functions Under the Windows System. 1.Set up triggers 2.ISP Tuning 3.Read lens temperature	2025/10

Appendix

I. Here's a Python example for converting UYVY to YUYV:

1.1. Implementation Approach

Understanding YUV422 UYVY and YUYV Formats:

UYVY Format: Two adjacent pixels share a pair of UV chrominance components.

The byte layout is U0 Y0 V0 Y1.

YUYV Format: Also shares UV components between two pixels, but ordered as Y0 U0 Y1 V0.

Data Conversion:: Conversion from UYVY to YUYV requires only reordering the bytes without altering the actual pixel data.

1.2. Python Example Code (For Reference Only):

```
import cv2

import numpy as np

def uyvy_to_yuyv(uyvy_data, width, height):
    # Convert input UYVY data to a numpy array
    uyvy_array = np.frombuffer(uyvy_data, dtype=np.uint8)
    # Reshape the array to match image dimensions
    uyvy_array = uyvy_array.reshape((height, width * 2))
    # Create an empty array for YUYV
    yuyv_array = np.empty_like(uyvy_array)
    # Process each row
    for i in range(height):
        for j in range(0, width * 2, 4):
            # Reorder bytes from UYVY to YUYV
            yuyv_array[i, j] = uyvy_array[i, j + 1]
            yuyv_array[i, j + 1] = uyvy_array[i, j]
            yuyv_array[i, j + 2] = uyvy_array[i, j + 3]
            yuyv_array[i, j + 3] = uyvy_array[i, j + 2]
    # Convert the YUYV array back to bytes
    yuyv_data = yuyv_array.tobytes()
```



```

    return yuyv_data

# Example Usage

width = 640
height = 480

# Generate random UYVY data for demonstration
uyvy_data = np.random.randint(0, 256, width * height * 2, dtype=np.uint8).tobytes()

# Convert UYVY to YUYV
yuyv_data = uyvy_to_yuyv(uyvy_data, width, height)

# Convert YUYV data to an OpenCV image
yuyv_array = np.frombuffer(yuyv_data, dtype=np.uint8).reshape((height, width * 2))

# Convert YUYV format to BGR for display in OpenCV
bgr_image = cv2.cvtColor(yuyv_array, cv2.COLOR_YUV2BGR_YUYV)

# Display the image
cv2.imshow('YUYV Image', bgr_image)

cv2.waitKey(0)

cv2.destroyAllWindows()

```

1.3.Code Explanation:

A.Conversion Function: `uyvy_to_yuyv` takes UYVY byte data, image width, and height as inputs, returning YUYV-formatted byte data.

B.Data Reordering: Uses nested loops to process each 4-byte block, reordering bytes from UYVY to YUYV layout.。

C.Example Workflow: Generates synthetic UYVY data, converts it to YUYV, then visualizes the result using OpenCV.

D.Image Display: Converts YUYV data to BGR format (compatible with OpenCV) for visualization.

E.Notes

a.Data Integrity: Ensure the input UYVY data length matches the specified width and height (total bytes = width * height * 2).

b.Real-World Usage: Replace the random test data with actual UYVY streams from camera devices or video sources.

II. Python Example for YUYV to UYVY Conversion

2.1. Implementation Approach

1.1.1 Characteristics of YUYV and UYVY Formats

YUYV Format: Two adjacent pixels share a set of UV components, with data arranged as Y0 U0 Y1 V0.

UYVY Format: Similarly, two adjacent pixels share a set of UV components, but the arrangement is U0 Y0 V0 Y1.

1.1.2 Conversion Method: Essentially, it involves reordering the data—restructuring the data from the YUYV format to follow the UYVY sequence.

2.2. Python Example Code (For Reference Only)

```
import cv2

import numpy as np

def yuyv_to_uyvy(yuyv_data, width, height):
    # Convert input YUYV data to a numpy array
    yuyv_array = np.frombuffer(yuyv_data, dtype=np.uint8)

    # Reshape the array to match the image dimensions
    yuyv_array = yuyv_array.reshape((height, width * 2))

    # Create an empty array for storing UYVY data
    uyvy_array = np.empty_like(yuyv_array)

    # Process data row by row
    for i in range(height):
        for j in range(0, width * 2, 4):
            # Reorder data to follow the UYVY sequence
            uyvy_array[i, j] = yuyv_array[i, j + 1]
            uyvy_array[i, j + 1] = yuyv_array[i, j]
            uyvy_array[i, j + 2] = yuyv_array[i, j + 3]
            uyvy_array[i, j + 3] = yuyv_array[i, j + 2]

    # Convert the UYVY array back to byte data
    uyvy_data = uyvy_array.tobytes()

    return uyvy_data
```

```
# Example Usage
```

```
width = 640
```

```
height = 480
```

```
# Generate random YUYV data for demonstration
```

```
yuyv_data = np.random.randint(0, 256, width * height * 2, dtype=np.uint8).tobytes()
```

```
# Call the conversion function
```

```
uyvy_data = yuyv_to_uyvy(yuyv_data, width, height)
```

```
# Convert UYVY data to an OpenCV image
```

```
uyvy_array = np.frombuffer(uyvy_data, dtype=np.uint8).reshape((height, width * 2))
```

```
# Convert the UYVY format image to BGR format for display in OpenCV
```

```
bgr_image = cv2.cvtColor(uyvy_array, cv2.COLOR_YUV2BGR_UYVY)
```

```
# Display the image
```

```
cv2.imshow('UYVY Image', bgr_image)
```

```
cv2.waitKey(0)
```

```
cv2.destroyAllWindows()
```

2.3.Code Explanation

2.3.1.Function `yuyv_to_uyvy`: This function takes YUYV data, image width, and height as inputs.

A. First, it converts the input YUYV data into a numpy array and reshapes it to match the image dimensions.

B.It creates an empty array `uyvy_array` with the same size as `yuyv_array` to store the converted data.

C.Using nested loops, it processes data row by row, reassigning YUYV data to `uyvy_array` following the UYVY arrangement rules.

D.Finally, it converts the processed `uyvy_array` back to byte data and returns it.

2.3.2. Example Usage:

A.Sets the image width and height to 640 and 480, respectively.

B.Generates random YUYV data as sample input.

C.Calls the `yuyv_to_uyvy` function for format conversion.

D.Converts the resulting UYVY data into a numpy array and uses `cv2.cvtColor` to convert

it from UYVY to BGR format, enabling display with `cv2.imshow.`

E.Uses `cv2.waitKey(0)` to wait for user input and `cv2.destroyAllWindows()` to close all open windows.

2.4. Notes

A.Ensure the length of the input YUYV data matches the specified image width and height.

B.The random YUYV data generated in the code is for demonstration only; real YUYV data should be used in practical applications.